A Separation Principle for Lookup-Based zkML: Activation-Function Structure Cannot Reduce Per-Lookup Proving Cost

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Abstract

In zero-knowledge machine learning (zkML), the dominant cost is generating the proof, not running the model, and it concentrates in the nonlinearities a transformer must evaluate inside the proof system. It is tempting to exploit a nonlinearity's mathematical structure (low degree, parity, or kernel form) to prove it more cheaply. We show this hope is misplaced for the dominant cost: in a Shout-style (one-hot) lookup argument the per-lookup proving work is a function of the access pattern alone, never of the table values, so function structure has *zero* leverage on it. This is a **separation principle**; structure can cheapen only a secondary, once-per-proof table term. That table term stays subordinate as models get deeper because the only unbounded data-dependent amplifier of per-layer error in a pre-LN transformer is the LayerNorm gain $1/\sigma$: a σ-floor on typical inputs, with the measured σ-growth above it, lets a single fixed proving precision suffice at every depth, keeping proof cost near-linear in the number of layers. Whether a fixed precision survives depth is a property of the network, shared by FHE and integer-quantized inference; what is paradigm-specific is the pricing: the leverage of structure inverts between FHE's multiplicative-depth metric and the lookup address metric. We measure this depth-to-cost scaling on two independent proving systems (EZKL/halo2 and Jolt Atlas), and turn the one dial the separation leaves open, the committed address width, into a bit-exact, upstreamed reduction in prover time.

1 Introduction

Verifiable inference is becoming a deployment requirement: a service that runs a model on someone's behalf (an outsourced LLM, a private on-chain model) may be asked to prove it ran the computation it claims. Zero-knowledge proofs make this possible, but at a price. Generating the proof, not running the model, is the dominant cost, and it typically sits orders of magnitude above native inference. That cost concentrates in the nonlinearities: a transformer's softmax must be evaluated inside the proof system, where exp is approximated by a lookup of fidelity r bits. More fidelity means smaller approximation error but a larger, costlier table.

The proving cost of a compiled nonlinearity splits into a per-proof table term (paid once, at fidelity r) and a per-lookup address term (paid m times), and two questions decide the total. First, with r fixed, can the function's own structure cheapen the proof? On a Shout-style one-hot lookup backend the answer is a sharp no: a **separation principle** (§9) shows the per-lookup cost is a

*This work was carried out independently, in the author's own time and personal capacity.

function of the access multiset alone, never of the table values, so function structure (the whole cost under FHE's multiplicative-depth metric) has *zero* leverage on the per-lookup term, and the leverage of structure inverts between the two paradigms. Second, must the fidelity r grow with the depth L? This is the question every deployed zkML system answers implicitly, and the stakes are binary: if r must grow, the per-proof term overtakes the per-lookup one and deep models are out of reach; if a fixed precision suffices, the fixed-point quantization these systems already choose is licensed rather than lucky. We show a fixed precision does suffice — this paper supplies that license — and that is what keeps the address term (the one structure cannot touch) dominant.

The licensing question is not special to zkML: whether a fixed precision survives depth is a property of the network, so any fixed-precision deployment (FHE, integer quantization) faces it, and what differs across paradigms is how that precision is priced, which is what we work out and measure for the zkML lookup setting (§9). That the normalized forward error stays bounded with depth is corroborated independently outside zkML, in a recent low-precision numerical-stability theory [4] and in post-training quantization [19]; what those settings lack, and we supply, is the pricing of that bounded error as proof cost. The pivot is a single observation: in a pre-LN transformer the only unbounded data-dependent amplifier of a layer's error is the LayerNorm gain $1/\sigma$, so the entire depth-scaling question collapses onto one statistical object: the distribution of the residual scale σ across layers. Control σ and you control the proof cost.

This trade is worth setting up carefully, because the rest of the paper lives inside it. Approximating layer ℓ's nonlinearity to r_ℓ bits gives local error $\varepsilon_\ell = C\,2^{-r_\ell}$: every added bit halves the local error and (for a dense table) doubles the size of the table that must be proved. Local errors do not reach the output equally — an error injected early is amplified by everything downstream of it — and we write S_ℓ for that *sensitivity*: how much one unit of layer-ℓ error moves the output. Each table of fidelity r costs $\tilde{K}(r)$, where the *cost kernel* $\tilde{K}$ is a property of the proving backend, not of the model. The prover's whole game is then a constrained allocation of the fidelities $\{r_\ell\}$:

$$\min_{\{r_\ell\}} \sum_\ell \tilde{K}(r_\ell) \qquad \text{subject to} \qquad \sum_\ell S_\ell\,\varepsilon_\ell \le E^*,$$

i.e. spend bits where the output is most sensitive, subject to a correctness budget E^* on the output. The sensitivities S_ℓ are set by the network and the kernel $\tilde{K}$ by the backend; the next two paragraphs pin down the first, and §5 prices the second.

Where can S_ℓ blow up? In a pre-LN transformer a layer's contribution passes through LayerNorm, and LayerNorm divides by the residual scale σ: when the residual is small, the normalization *magnifies* whatever error rides on it. This gain is the only unbounded data-dependent amplification of layer error in the network. We prove (Lemma 1) that the per-layer error-propagation operator obeys $\|B_k\| \le 1 + c_k/\sigma_k$, where $c_k = \|J_{\mathrm{Block},k}\|\,\|\gamma_k\|_\infty$ collects the a-priori-bounded part — the block's Jacobian norm times the LayerNorm weight scale. Chaining the bound through every layer above ℓ gives

$$S_\ell \;\le\; \prod_{k>\ell} (1 + c_k/\sigma_k).$$

The numerator of the gain is measured, not assumed: c_k is profiled per layer on four models spanning both normalization families, and it increases with depth — layer- and model-dependent, not a universal $O(1)$ constant (per-model values, medians, and the attention-block operator-norm convention are collected in Lemma 1 and §10). The unbounded factor is the denominator. Everything therefore reduces to the distribution of the residual scale $\sigma_k(x) = \|Px_k\|/\sqrt{d}$, with $P = I - \mathbf{1}\mathbf{1}^\top/d$ the mean-removal projector: this one per-layer statistic decides whether the sensitivities — and with them the whole allocation — stay tame.

Which regime that distribution places us in turns on the threat model. Whether the input is typical ($x \sim \mathcal{D}$, honest-input zkML) or adversarial ($x \in A(X)$, DoS / public verification) determines the regime. The dial is the cancellation coefficient η: typical inputs give $\eta < 1$ (blocks do not systematically cancel the residual), adversaries can drive $\eta \to 1$ (collapse). The adversary here threatens proving cost and availability (DoS, or the predictability of a public-verification cost SLA), not soundness: ZK already guarantees the output cannot be misreported regardless of σ; our regimes concern how expensive an honest proof is, not whether a dishonest one could pass.

Our results populate this axis, in three moves. The first move closes the average-case regime: under $\mathcal{D}$, the σ-floor splits into its two mechanisms. A deterministic *mean-spread floor* keeps σ away from zero using only the mean component of the residual (Proposition 1, unconditional given two checkable assumptions), and an *alignment-driven growth* explains why σ in fact rises with depth: successive residual increments share a common direction, and alignment forces growth (Theorem 1). The growth theorem is confirmed on real models directly from the mean-activation vectors, not through a proxy: the growth condition ($\theta_k < 0$) holds in 24/24 blocks on GPT-2 with growth exponent $p \approx 2$, and the exponent is re-measured on ten models (two gated) at or above the linear-growth boundary (§8.2, EXP-1R/EXP-A).

The second move prices the floor. Since the cost kernel is convex in r, the optimal allocation is water-filling ($r_\ell^* \propto \log_2 S_\ell$): fidelity concentrates where residuals are small and sensitivities large. For a Shout-style backend the dominant per-lookup address commit is r-independent, giving proof cost $\approx O(L)$; a boundary lemma closes the $\Theta(L)$ total exactly at the measured $\alpha = 1$ growth boundary, so no unproven $\alpha > 1$ is needed (EXP-A, Lemma 5). We then measure the depth→cost scaling itself on two independent proving systems — circuit work scales $L^{1.05}$ on EZKL/halo2, and Jolt prover time is affine in depth ($R^2 = 0.999$) — and, probing the kernel directly, we show the floor is load-bearing: a σ-collapsed model would demand per-layer tables far past the build wall, while honest inputs sit at the deployed fidelity (§8.5).

The third move is the structure axis. The separation principle is not only proved but measured, on two falsification channels (§9); and its one remaining dial — the committed address width — becomes a deployed win: removing 5 of the 16 committed one-hot polynomials per softmax node (the $\propto N_{\text{seq}}^2$ axis) cuts prover time 2.6–6.0% on the four PR models, bit-exact and upstreamed as a public PR.

How to read the paper. The theoretical spine is §2–§5: setup and the regime split, the floor-and-growth mechanism, and its allocation/cost consequences. §6 and §7 bound what an adversary can do to σ and explain where the mechanism's premise (massive activations) comes from; the measurements are collected in §8. The structure axis — the separation principle and the activation classification — is §9 and can be read independently; §10 maps every claim to its assumptions, and proofs are in Appendix A.

Scope and contingency. Two limits are load-bearing and stated up front. First, the average-case regime is contingent: it rests on massive activations being input-robust, a training-induced and architecture-dependent property rather than a theorem, so sink-free designs (Softpick, gated attention) that remove the attention sink also remove the super-linear σ-growth and the Region 2.5 defense (§7). Second, the structural lower bounds are restricted-model and content-agnostic: rank/degree bounds within a fixed prover class, with the top-k selection escape left open (§9), not unconditional complexity bounds.

Contributions. The most distinctive results are the separation principle and the content-agnostic attention rank gap; the σ-floor and its two-system cost measurement are what place proving in the address-dominated regime where they bite.

1. A *cost-polynomiality target* (σ-floor $\wedge$ bit-polynomial kernel) and a worst/average threat-model split (§3).
2. A *mean/fluctuation decomposition* $\sigma_k^2 = m_k^2 + f_k^2$ and a deterministic mean-spread floor (Proposition 1, §4) that holds even when fluctuation concentration fails.
3. A *Gram-geometric growth theorem* (Theorem 1, §4) linking depthwise σ-growth to activation alignment.
4. *Water-filling allocation* and, for Shout (Jolt Atlas), $O(L)$ proof cost via r-independent address-commit dominance (§5), closed at the $\alpha = 1$ growth boundary with no unproven $\alpha > 1$ (Lemma 5; exponent measured cross-model, EXP-A) and measured on two independent ZK systems (§8.3).
5. A *partial adversarial robustness* result (Region 2.5, §6) and a structural justification of its premise via massive-activation origin (§7).
6. Real-model confirmation on GPT-2/medium (§8): the σ-floor, Theorem 1 directly from mean-activation vectors, and a real-model adversarial test of Region 2.5; plus an explicit assumption map (§10).
7. A *degree/rank-gap characterization of the attention N_{seq}^2 limit* (Theorem 3, Corollary 2, §8.5): every content-agnostic method is a rank-m kernel approximation needing degree $\propto R$ (residual range): measured directly, $\text{rank}_\delta(K) = \Theta(N_{\text{seq}})$ already forces $m = \Omega(N_{\text{seq}})$, and in the spherical-harmonic model (matched-scaling) $m \geq F_\delta \gg N_{\text{seq}}$; yet it can afford only $\propto \log N_{\text{seq}}$. A dot-product/spherical-harmonic argument makes polynomial features optimal in that model, so the bound covers random features too. This isolates the only escapes (content selection $\to$ soundness, model change $\to$ function) and organizes existing attention provers into three orthogonal cost axes: count, fidelity, and gauge (§11); a fine-grained complexity complement transplants the Alman–Song bounded-entry transition [2] to the prover side (completeness-stage inheritance, checkable threshold).
8. An *activation-axis classification and separation principle* (§9): differential-algebraic structure decides which cost regime an activation can reach (all transformer activations constructible; dense-forced only if differentially transcendental, the converse being gated by an open on-domain incompressibility frontier), while a Shout-relative separation principle proves function structure cannot touch the dominant per-lookup address term (dual measured falsification channels); the leverage of structure inverts between FHE's depth metric and lookup-SNARK address metrics. The remaining address-width dial yields a measured, deployed win: -5 committed polynomials per softmax node ($\propto N_{\text{seq}}^2$ axis), prover -2.6–-6.0% on the four PR models (-0.2–-11.3% on three further in-tree exports; the invariant is -5 polynomials per node, reproduced exactly on seven models, the percentage is the softmax commitment share), bit-exact, soundness-preserving, upstreamed as a public PR.

2 Preliminaries

Residual stream $x_{k+1} = x_k + g_k$, $g_k = \text{Block}_k(\text{LN}(x_k))$ (attention or MLP). Distribution $\mathcal{D}$ over inputs; all expectations below are assumed finite (activations square-integrable under $\mathcal{D}$). Write

$\mu_k = \mathbb{E}_{\mathcal{D}}[x_k]$, $\xi_k = x_k - \mu_k$. Define

$$s_k^2 = \sigma_k^2 := \mathbb{E}_{\mathcal{D}} \|Px_k\|^2 / d \qquad \text{(mean-square residual scale)} \tag{1}$$

$$\sigma_{g,k}^2 := \mathbb{E}_{\mathcal{D}} \|Pg_k\|^2 / d \qquad \text{(block injection energy)} \tag{2}$$

$$\eta_k := -\mathbb{E}_{\mathcal{D}} \langle Px_k, Pg_k \rangle / (d\, s_k\, \sigma_{g,k}) \in [-1, 1] \qquad \text{(cancellation; Cauchy–Schwarz)} \tag{3}$$

Lemma 1 (amplification)**.** $\|J_{\mathrm{LN}}\| \le \|\gamma\|_\infty / \sigma$*, hence* $\|B_k\| \le 1 + c_k/\sigma_k$ *with* $c_k = \|J_{\mathrm{Block},k}\|\, \|\gamma_k\|_\infty$ *bounded a priori; the only unbounded data-dependent amplifier is* $1/\sigma_k$*.* (Proof in Appendix A.) *(Measured, EXP-3, per-token diag convention, four models across two normalization families:* $c_k \in [0.17, 274]$ *median* 19 *(GPT-2-medium),* $[3.3, 101]$ *median* 12 *(GPT-2-small) on LayerNorm;* $[0.08, 23]$ *median* 1.8 *(Qwen2.5-0.5B),* $[0.04, 3.6]$ *median* 0.3 *(TinyLlama-1.1B) on RMSNorm; all depth-increasing. For attention blocks the operator norm entering the* S_ℓ *product is the* seq-opnorm *figure — up to* 640 *(medium) /* 133 *(small), §10 — the per-token diagonal understates token-mixing; with either convention* c_k *is a priori bounded.)*

Energy recursion. Expanding $\mathbb{E}\|Px_{k+1}\|^2$:

$$s_{k+1}^2 = (s_k - \eta_k \sigma_{g,k})^2 + (1 - \eta_k^2)\, \sigma_{g,k}^2. \tag{$\star$}$$

3 Three Regimes

Regime	Threat	Result	Status
1. Low-rank residual	adversarial (unconditional)	poly(L)	proved; narrow
2. Full-width worst-case	adversarial	**impossible** (S exp.)	closed (impossibility)
2.5. Full-width, weak adv.	cannot suppress massive acts.	partial robustness	this work (cond.)
3. Full-width average-case	typical $x \sim \mathcal{D}$	poly(L); $O(L)$ on Jolt (bounded r_ℓ or Lemma 5; §10)	**closed** (avg-case, this work)

Regime 2's impossibility: an adversary that can drive $\sigma_k(x)$ small on alternating layers makes S_ℓ exponential — and a strong-budget adversary measurably can (§8.4); §6 is what confines weaker ones. Under $\mathcal{D}$, however, such trajectories are exponentially rare (measure $\sim e^{-\Theta(d)}$ under standard full-width concentration, stated heuristically). The practical setting (full-width, standard LoRA, honest-input) lives in Regime 3.

4 The σ-Floor (Regime 3, sensitivity L)

4.1 Orthogonal decomposition

Since $\mathbb{E}[\xi_k] = 0$, the cross term vanishes:

$$\sigma_k^2 = \|P\mu_k\|^2 / d + \mathbb{E}\|P\xi_k\|^2 / d \;=:\; m_k^2 + f_k^2. \tag{1}$$

m_k^2 (mean-spread) is **deterministic** (a property of the distribution's mean, input-independent); f_k^2 is the fluctuation variance. Empirically m_k^2 is ≈50–84% of σ_k^2 through most of the depth, dropping to ≈30% in the first embedding-transient layers (§8).

4.2 The deterministic mean-spread floor (Proposition 1)

With $\gamma_k = \mathbb{E}_{\mathcal{D}}[g_k]$, $\mu_{k+1} = \mu_k + \gamma_k$, the mean obeys a recursion isomorphic to ($\star$):

$$m_{k+1}^2 = (m_k - \theta_k \nu_k)^2 + (1 - \theta_k^2)\, \nu_k^2, \quad \nu_k^2 = \|P\gamma_k\|^2 / d, \;\; \theta_k = -\langle P\mu_k, P\gamma_k\rangle / (d\, m_k \nu_k). \qquad (\star\star)$$

Proposition 1 (deterministic mean-spread floor)**.** *If $\theta_k \le \theta^* < 1$ and $\nu_k \ge \nu_{\min} > 0$ for all k, then $m_k \ge \sqrt{1-\theta^{*2}} \cdot \nu_{\min} > 0$ for all $k \ge 1$ (and for $k = 0$ as well whenever $m_0 \ge \sqrt{1-\theta^{*2}}\, \nu_{\min}$).*

Proof. Write $B = \sqrt{1-\theta^{*2}}\, \nu_{\min}$, and recall $|\theta_k| \le 1$ (Cauchy–Schwarz). *Base case ($k = 1$), any $m_0 \ge 0$:* if $m_0 = 0$ then $P\mu_0 = 0$, so the cross term $\langle P\mu_0, P\gamma_0\rangle$ (i.e. $\theta_0 m_0 \nu_0$) vanishes and $m_1^2 = \nu_0^2 \ge \nu_{\min}^2 \ge B^2$ directly, with θ_0 immaterial. For $m_0 > 0$ (so that θ_0 is well defined): if $\theta_0 \ge 0$ then $m_1^2 \ge (1-\theta_0^2)\nu_0^2 \ge (1-\theta^{*2})\nu_{\min}^2 = B^2$; if $\theta_0 < 0$ then $m_1^2 = (m_0 - \theta_0\nu_0)^2 + (1-\theta_0^2)\nu_0^2 \ge \theta_0^2\nu_0^2 + (1-\theta_0^2)\nu_0^2 = \nu_0^2 \ge \nu_{\min}^2 \ge B^2$; so $m_1 \ge B$. *Inductive step ($m_k \ge B \Rightarrow m_{k+1} \ge B$):* from ($\star\star$) (here $m_k \ge B > 0$, so θ_k is defined), if $\theta_k \ge 0$ then $m_{k+1}^2 \ge (1-\theta_k^2)\nu_k^2 \ge (1-\theta^{*2})\nu_{\min}^2 = B^2$ (reset, independent of m_k); if $\theta_k < 0$ then $-2\theta_k\nu_k m_k \ge 0$, so $m_{k+1}^2 = m_k^2 - 2\theta_k\nu_k m_k + \nu_k^2 \ge m_k^2 \ge B^2$ (growth). Hence $m_k \ge B$ for all $k \ge 1$. □

Because m_k is deterministic, this floor survives even when fluctuation concentration (the Hanson–Wright argument on f_k) is vacuous, which is the empirically relevant case (the effective rank of the fluctuation covariance is $O(1)$, not $\Theta(d)$, in real models). Thus $\sigma_k^2 \ge m_k^2 \ge \text{floor}^2$.

Remark 1 (the cancellation coefficient splits by channel)**.** With $\delta_k = g_k - \gamma_k$, fluctuation injection $\sigma_{\delta,k}^2 = \mathbb{E}\,\|P\delta_k\|^2 / d$, and fluctuation cancellation $\varphi_k = -\mathbb{E}\langle P\xi_k, P\delta_k\rangle / (d\, f_k\, \sigma_{\delta,k})$, the zero-mean cross terms vanish, so $\mathbb{E}\langle Px_k, Pg_k\rangle = \langle P\mu_k, P\gamma_k\rangle + \mathbb{E}\langle P\xi_k, P\delta_k\rangle$ and, with $s_k^2 = m_k^2 + f_k^2$ and $\sigma_{g,k}^2 = \nu_k^2 + \sigma_{\delta,k}^2$, $\eta_k = w_1\theta_k + w_2\varphi_k$ with $w_1 = m_k\nu_k/(s_k\sigma_{g,k})$ and $w_2 = f_k\sigma_{\delta,k}/(s_k\sigma_{g,k})$. Cauchy–Schwarz on (m_k, f_k) and $(\nu_k, \sigma_{\delta,k})$ gives $w_1 + w_2 \le 1$ (both nonnegative), whence $\eta_k \le \max(\theta_k, \varphi_k)$ wherever that maximum is nonnegative. Bounding both channel cancellations below 1 therefore bounds η_k below 1: a σ-collapse needs the mean or the fluctuation channel to cancel, not merely their aggregate. (The η-decomposition is among the identities verified numerically, §8.)

4.3 Alignment ⇒ growth (Theorem 1)

With $P\mu_0 = 0$, $P\mu_k = \sum_{j<k} P\gamma_j$, so

$$\langle P\mu_k, P\gamma_k\rangle = \sum_{j<k} \langle P\gamma_j, P\gamma_k\rangle = \sum_{j<k} G_{jk} \quad \text{(Gram matrix), and} \quad \theta_k \le 0 \iff \sum_{j<k} G_{jk} \ge 0.$$

Theorem 1 (alignment ⇒ growth)**.** *Write $P\gamma_j = a_j u + v_j^\perp$ for a fixed unit vector u (so $a_j = \langle P\gamma_j, u\rangle$ and $v_j^\perp \perp u$). If* (H1) *$a_j \ge 0$ $\forall j$ (common half-space) and* (H2) *$\sum_{j<k} a_j a_k \ge \left|\sum_{j<k}\langle v_j^\perp, v_k^\perp\rangle\right|$, then $\theta_k \le 0$ (growth). With $\nu_k = \Theta(1)$: under full alignment $m_k^2 = \Theta(k^2)$; under no alignment $m_k^2 = \Theta(k)$ (random walk).* (Proof in Appendix A.)

The common direction u is a massive-activation coordinate; the alignment is structurally induced — a training incentive, not a theorem — by the residual stream's additivity (§7). This makes Regime-3's growth mechanism the **mean-spread**, not the effective rank.

Corollary 1 (depth-bounded normalized discrepancy; LoaQ as a symptom)**.** *In the average-case regime (§3), let E_k be the accumulated pre-normalization error at layer k under a fixed per-layer precision ε, so that $E_k \le \sum_{\ell \le k} \prod_{\ell < j \le k}(1 + c_j/\sigma_j)\, \varepsilon$ (Lemma 1). Assume the growth of Theorem 1 holds at its full-alignment rate, $\sigma_k \ge a\, k$ for $k \ge k_0$ (equivalently $m_k = \Theta(k)$): the $p \approx 2$ regime*

measured in §8.2), with $\sigma_k \geq \sigma_{\min}$ elsewhere (Proposition 1), $\bar{c} = \sup_k c_k$ (seq-opnorm convention on attention blocks, §10) and $\sup_k \|\gamma_k\|_\infty$ finite (frozen parameters; Lemma 1), and $\beta = \bar{c}/a < 1$. Then $\prod_{\ell<j\leq k}(1 + c_j/\sigma_j) \leq C_0\,(k/\ell)^\beta$ with $C_0 = (1 + \bar{c}/\sigma_{\min})^{k_0}$, so $E_k \leq C_0\,\varepsilon\,k/(1-\beta) + O(k^\beta) = O(k)$, and the discrepancy between the exact and the fixed-precision normalized sub-block outputs, at most $(\|\gamma_k\|_\infty/\sigma_k)\,E_k$ by Lemma 1, is bounded in depth; under super-linear growth ($\sigma_k/k \to \infty$) the bound eventually decreases to zero. The floor alone does not suffice: with no amplification at all ($c \equiv 0$), $E_k = k\varepsilon$ still grows linearly, and it is the σ-growth that caps the normalized ratio. This depth-boundedness of the normalized discrepancy is exactly what LoaQ [19] reports empirically for RMSNorm models; growth, with the floor beneath it, is the mechanism.

5 Allocation and Cost

Guarantee. Quantization robustness is a triangle inequality: $\sigma_k \geq \sigma_{\text{committed}} - \varepsilon/\sqrt{d}$; the witness contains σ_k, so the floor is checkable in-circuit.

Allocation. Solve $\min \sum_\ell \tilde{K}(r_\ell)$ s.t. $\sum_\ell S_\ell\, C\, 2^{-r_\ell} \leq E^*$. Throughout, E^* is a *relative* budget, pinned to the output scale ($E^* = \Theta(\sigma_L)$: fixed relative accuracy); under this convention the water-filling level A of Lemma 3 stays bounded above in L for $\alpha \geq 1$ (an absolute, L-independent E^* would instead give $A = \log_2(CL/E^*) = \Theta(\log L)$ on the bit-as-cost kernel, inflating every layer's allocation), and the L-asymptotics of Lemma 4 and Theorem 2 are stated in it. The KKT stationarity $\tilde{K}'(r_\ell) = \lambda \ln 2 \cdot S_\ell \varepsilon_\ell$ gives, for **convex, differentiable, strictly increasing** $\tilde{K}$, the **water-filling** solution (Lemma 3)

$$r_\ell^* = \tfrac{c}{c+1}\,(\log_2 S_\ell + \text{const}) \qquad : \text{high fidelity early (small } \sigma\text{, large } S\text{), low fidelity deep (large } \sigma\text{).}$$

Kernel. The backend we measure for $O(L)$ (Jolt Atlas) uses **Shout** (Twist & Shout, ePrint 2025/105) [25], not Lasso/Surge; we extracted its prover-cost accounting from the source (EXP-4d Stage 0). Per nonlinear op, Shout charges an r-**independent** per-lookup one-hot **address commit** ($\sim d\,m$ field elements, the unique dominant m-term) plus an r-**dependent** per-proof **table-MLE evaluation** (dense $O(2^r)$, or prefix-suffix $c\,2^{r/c}$, degree 2), with **no** grand-product / offline-memory-checking tax (absent from the source). Writing the node cost (whose r-dependent, per-proof part is the table term) as

$$\tilde{K}(r) = \kappa_{\text{addr}}\, d\, m + \kappa\, c\, 2^{r/c} \qquad (\text{dense is } c = 1,\ 2^r).$$

The r-dependent term is **convex in** r, so the allocation is water-filling, not bang-bang. (Sublinearity in table size n must not be confused with concavity in the allocation variable r: $n = 2^r$ flips n-concavity to r-convexity. Lasso's table term $c\,n^{1/c}$ has the same r-shape, which is why a Lasso-framed kernel reproduces the right scaling while mis-describing the mechanism.)

Theorem 2 (end-to-end cost)**.** *If $\sigma_k \sim \sigma_0 k^\alpha$ (growth) and $c_k = \Theta(1)$ (bounded above and below across depth, as in Lemma 4; measured, EXP-3), then, under the relative-budget convention $E^* = \Theta(\sigma_L)$ of §5:*

- *bit-as-cost backends: $\sum r_\ell^* = O(L^{\max(1,\,2-\alpha)})$, and $\Theta(L^{2-\alpha})$ for $\alpha \leq 1$ (down from L^2 at $\alpha = 0$);*
- ***Shout (Jolt Atlas):** the per-lookup address commit $\sim d\,m$ is r-independent and $\propto$ operations, so summed over depth it is $O(L)$ regardless of fidelity; the only r-dependent cost is the per-proof table-MLE eval $\sim 2^{r_\ell}$. Under a linear floor with $\beta = \bar{c}/a < 1$ — the hypotheses of Corollary 1,*

with $\alpha \geq 1$ now measured cross-model (EXP-A, §8.2) — the summed table term is $O(L)$ even at the boundary $\alpha = 1$ (Lemma 5), so the total is $\Theta(L)$ with no per-layer boundedness (no $\alpha > 1$) needed; at the deployed fixed fidelity the same conclusion is the Regime-3 reading (the σ-floor of Prop. 1/Thm 1 rules out the collapse that would force r_ℓ up, and Corollary 1 keeps the fixed fidelity sufficient).

(Proof in Appendix A.)

A summed table term $O(L)$ therefore gives $\Theta(L)$; σ-collapse (Regime 2) drives r_ℓ up, making the per-proof 2^{r_ℓ} explode into a build wall (EXP-4a): the Shout-language form of the cost catastrophe. So the floor-and-growth mechanism bounds the per-proof table term: the floor rules out the collapse that would drive $r_\ell = \log_2 S_\ell + A$ past the wall, and the measured growth (deep-half $\alpha \geq 1.46$ on 10/10 models, bulk $\alpha \geq 0.9$ on 8/10, EXP-A) closes the summed term at the $\alpha = 1$ boundary (Lemma 5) while holding the near-optimal r_ℓ near the deployed fidelity; the proof cost is then near-linear in the number of operations. This Shout $O(L)$ is a Regime-3 statement: it holds at the deployed fixed fidelity, which the σ-floor keeps sufficient by ruling out the σ-collapse that would force r_ℓ upward; the governing assumptions are collected in the map of §10.

Measured (§8.3). The depth→cost prediction is confirmed on two independent backends: EZKL (halo2) circuit work scales $L^{1.05}$ and Jolt Atlas (Twist & Shout) prover time is affine in depth ($13.98\,L + 6.67$ ms, $R^2 = 0.999$), i.e. $L^{0.95}$. Within the measured fidelity range the per-proof table term 2^r is not yet triggered, consistent with the address-commit-dominated regime. We measure the cost kernel itself and the σ-floor's load-bearing role separately (EXP-4a/4b, §8.5).

6 Partial Adversarial Robustness (Region 2.5)

For an individual input x, the triangle inequality gives

$$\sigma_k(x) = \|P\mu_k + P\xi_k(x)\| / \sqrt{d} \geq m_k - \|P\xi_k(x)\| / \sqrt{d}. \tag{2}$$

To collapse $\sigma_k(x)$, an adversary must cancel the deterministic mean $P\mu_k$ with its (input-dependent) fluctuation. If the reachable fluctuation radius $\rho_k := \sup_{x \in A(X)} \|P\xi_k(x)\| / \sqrt{d} < m_k$, then $\sigma_k(x) \geq m_k - \rho_k > 0$ regardless of the adversary. Since m_k grows with depth (Thm 1), **shallow layers are adversarially vulnerable but deep layers become adversarially robust.** Because the exponential blow-up of $S_\ell = \prod(1 + \bar{c}/\sigma_k)$ requires simultaneous collapse across many layers (impossible along a single adversarial trajectory once deep layers are floored), the cost catastrophe is blocked.

This does **not** contradict Regime 2's impossibility; it conditions it. Regime-2 impossibility requires an adversary capable of **suppressing massive activations**. If massive activations are input-robust (§7), a weaker adversary is confined to Region 2.5. A real-model adversarial study on GPT-2 (§8.4) supports this depth spectrum and exhibits the predicted budget transition: a weak adversary ($\|\delta\| \leq 0.25\,\|\text{emb}\|$) leaves deep-layer massive coordinates 42–62% intact, while a stronger one suppresses them ($\leq 17\%$), recovering the Regime-2 collapse. We are honest that the support is qualitative: the achieved σ_{adv} stays **below** m_k (the floor inequality (2) holds only loosely, since the reachable radius ρ_k is large), and the final LayerNorm $\ln_f$ is the weakest point (massive survival $\leq 8\%$ at all budgets). Region 2.5 is thus a genuine but **budget-dependent** partial defense. (This concerns cost/DoS robustness, i.e. the expense and predictability of honest proving under adversarial inputs, not soundness, which ZK guarantees unconditionally.)

7 Origin of Massive Activations

Both Theorem 1 (alignment) and Region 2.5 (input-robust mean) rest on the existence and input-robustness of massive activations. The literature supplies a structural account:

- **Emergence:** massive activations emerge abruptly in early layers and persist at near-constant values along the residual stream [30]. The root cause is that softmax attention (weights summing to 1) has no "attend to nothing" option, so the model routes surplus probability mass through the first token (a "trash-can") under **causal masking** (the first token is never updated by other tokens, evolving only through its own self-attention and MLP path) [34, 12].
- **Input-robustness:** massive activations are **largely input-agnostic** (their values stay nearly constant across inputs [30]), and the sink sits at a fixed early position regardless of the initial token [34, 12]. The literature also **links attention sinks to slower perturbation propagation** (a robustness effect) [5], independently matching our Region 2.5.

Thus the alignment/robustness premises are structurally induced by softmax + causal masking, not arbitrary. **Caveat (honest):** this is a training incentive, not an unconditional theorem. Sink-free architectures (e.g. gated attention, Softpick [37]) remove massive activations, and with them the **super-linear growth** of Theorem 1 and the input-robust-mean premise of **Region 2.5**. They do **not**, by themselves, remove the existence of the mean-spread floor (Proposition 1), which requires only persistent, not-fully-anti-aligned mean-energy injection ($\nu_{\min} > 0$, $\theta^* < 1$); the depth-wise growth of hidden-state magnitude and variance that this premise needs is independently documented across large-scale Transformers (Peri-LN [16]). So what is genuinely softmax-specific is the rate of σ-growth and the adversarial defense, **not the floor's existence**. This contingency is measured (EXP-B19): on a matched Qwen3 softmax/gated pair, gating removes the massive activations (median max/median token-norm $35\times \to 1.9$–$2.4\times$) and the σ inflation (absolute deep σ, $38 \to 12$; per-step growth rate $\approx$ unchanged), while σ still grows, m_k still grows, and RMS $\geq \sigma$ on every layer, so the floor's existence and the $O(L)$ core survive; the floor is only weaker (worst-case amplifier 7.9–9.5 vs. 2.3). Proposition 1's premises, measured on GPT-2/LayerNorm, extend to four RMSNorm models (EXP-9, §10): the amplifier becomes 1/RMS with the identity $\mathrm{RMS}^2 = \sigma^2 + \mathrm{mean}^2 \geq \sigma^2$, so RMS $\geq \sigma$ on every layer and $S_\ell(\mathrm{RMSNorm}) \leq S_\ell(\mathrm{LN})$, with the residual norm growing $\times 80$–449 with depth; the gated pair (Gemma-2, Llama-3.2) is now measured on this amplifier axis as well (EXP-A, §8.2: RMS exponent = α_{bulk} to two decimals, at or above the linear boundary), complementing their attention-score geometry (EXP-B17, §8.5).

8 Experiments

Environment: PyTorch 2.12 (MPS), EZKL 23.0.5, Jolt Atlas (Rust, stable toolchain). σ-statistics via a hooked harness on residual-stream LayerNorm inputs; cost on real ZK provers via shared ONNX MLPs.

8.1 σ-floor and the mean-spread mechanism (GPT-2/medium, WikiText, 200k tokens)

- **σ monotone growth:** $0.17 \to 16.2$ (GPT-2-small), $0.096 \to 27.8$ (GPT-2-medium); no collapse. Cost proxy S_{typ} is 26–54 orders of magnitude below the collapse hypothesis $\Rightarrow$ polynomial.

- **Effective rank $O(1)$, not $\Theta(d)$** (median 1.2/1.3): the fluctuation-concentration path is vacuous; the mean-spread (≈50–84% of σ^2 over the bulk of the depth) carries the floor (Prop 1, not effective rank).
- $\eta^* = 0.42/0.05 < 1$**, mostly negative**; water-filling profile $r^*_\ell \approx$ 11–14 bits early vs. ≈4 bits deep, well below the build wall ≈ 24–26; end-to-end allocation cost 5–12× below the uniform-floor estimate.

8.2 Theorem 1 directly from mean-activation vectors (EXP-1R, EXP-A)

Re-running the harness to store the projected mean vectors $P\mu_k$ lets us compute the increment Gram $G_{jk} = \langle P\gamma_j, P\gamma_k\rangle$, the alignment axis, and the growth exponent directly (not via an inequality proxy):

	gpt2 (24 increments)	**gpt2-medium** (48)
$\theta_k < 0$ (growth)	**24/24**	44/48
$\theta_k < 0 \iff \sum_{j<k} G_{jk} > 0$ agreement	**23/23**	46/47
growth exponent p ($m_k^2 \sim k^p$)	**1.95**	2.02
Gram off-diagonal positive fraction	**1.00**	0.93
alignment explained variance	0.91	0.64

The increments concentrate on a **single common axis** (the massive-activation direction), giving $\theta_k < 0$ almost unconditionally and $m_k^2 \sim k^2$: a **direct confirmation of Theorem 1** (the $p \approx 2$ quadratic growth), cleaner than the inequality-proxy estimate (79%). Honest weak link: on medium the alignment is more diffuse (explained variance 0.64) with 4 early-layer exceptions.

Cross-model growth exponent (EXP-A). The σ-exponent itself, re-measured with a pre-registered bulk window ($k \geq \lceil L/4\rceil{+}1$, 95% CIs; deep half $k \geq \lceil L/2\rceil{+}1$; full range $k \geq 2$, the Fig. 3 convention) on ten models, 120k tokens each (gemma-2-2b: 40k, CPU):

Model	**norm**	L	α_{full}	$\alpha_{\text{bulk}}\pm$CI	α_{deep}
GPT-2	LN	12	0.90	1.19 ± 0.58	2.18
GPT-2-medium	LN	24	0.65	1.04 ± 0.33	2.04
Qwen2.5-0.5B	RMS	24	0.87	0.96 ± 0.26	1.88
Qwen2-0.5B	RMS	24	0.92	1.08 ± 0.26	1.97
TinyLlama-1.1B-Chat	RMS	22	1.42	1.92 ± 0.36	2.89
SmolLM2-360M	RMS	32	0.93	1.19 ± 0.27	2.28
Llama-3.2-1B (gated era)	RMS	16	0.94	1.10 ± 0.33	1.93
Qwen2.5-1.5B-Instruct	RMS	28	0.65	0.71 ± 0.22	1.46
Qwen2.5-Coder-1.5B-Inst.	RMS	28	0.67	0.79 ± 0.23	1.61
gemma-2-2b (gated)	RMS	26	0.88	1.55 ± 0.20	2.17

The log–log σ profile is convex (nested-window slopes non-decreasing on 10/10), so a single full-range fit is pulled down by the shallow transient: $\alpha_{\text{full}} \in [0.65, 1.42]$ is the $\alpha \approx 0.8$ regime of Fig. 3, a window artifact rather than a growth ceiling. In the deep half $\alpha \geq 1.46$ on 10/10 models (median ≈2.0); in the bulk window $\alpha \geq 0.9$ on 8/10 (the two Qwen-1.5B variants enter the bulk late, 0.71/0.79, with deep-half 1.46/1.61). The channel split matches Theorem 1's mechanism across families: $\alpha_m > \alpha_f$ on 10/10 ($m_k^2 \sim k^p$ with $p \in [1.6, 2.8]$, median ≈2.5, extending the $p \approx 2$ above to all ten), the fluctuation exponent is ≤0.15 on 7/10 (negative on five), and the RMS exponent

equals α_{bulk} to two decimals on every model, so the RMSNorm amplifier axis (EXP-9) inherits the same exponent — the two gated models close EXP-9's untested cell on that axis. Together this places the measured operating point at or above the $\alpha = 1$ boundary of Lemma 5, so the Shout $\Theta(L)$ no longer leans on the unproven $\alpha > 1$. Honest caveats: the exponents are window-dependent (hence the pre-registered windows and CIs), and gemma-2-2b ran at a third of the token budget on CPU.

8.3 Proof cost is $\approx O(L)$: measured on two ZK systems (EXP-2)

We export ReLU MLPs of depth $L \in \{2, 4, 8, 16, 32\}$ to ONNX and prove them on two backends:

System	Backend	depth → cost	proof size	bits → cost
EZKL 23.0.5	halo2 (KZG)	circuit num_rows $\sim L^{1.05}$	$\sim O(1)$ (20.1 KB)	scale 2–12 **flat**
Jolt Atlas	Twist & Shout	prove_ms $\sim L^{0.95}$, affine $13.98L{+}6.67$ (R^2=0.999)	$\sim O(L)$ (18→252 KB)	—

Both independent systems exhibit **depth→proof-cost** $\approx O(L)$, confirming the address-commit dominance of §5. Honest caveats: (i) this fixed-fidelity depth sweep does not exercise the table term (EZKL circuit size is flat across scale 2–12); we instead measure the cost kernel directly in EXP-4a/4b (§8.5); (ii) the clean cost signal is circuit work (num_rows), not wall-clock; (iii) num_lookups is not directly instrumented in Jolt Atlas, so prover time is the cost proxy; (iv) proof size differs by system ($O(1)$ on halo2 vs. $O(L)$ on Jolt Atlas); the prover time, our cost target, is $O(L)$ on both. These two experiments probe different axes: EXP-2 sweeps depth at fixed fidelity, whereas EXP-4a/4b exercise the fidelity/table term in isolation. A single depth-indexed sweep that lets the required fidelity grow with depth (EXP-B18) connects them: on the measured honest σ profile the per-layer fidelity increments decay with depth (as σ grows), so the compounded fidelity demand saturates and the address term stays dominant ($O(L)$), whereas a σ-collapse or a non-growing-σ profile drives the demand up linearly or super-linearly and triggers the 2^r term, reproducing EXP-4b (collapse: ≈85 bits, wall depth 6, cost ratio $\approx 10^{21}$). The robust content is this increment-decay shape; an absolute under-wall bit-count is convention-dependent, stated at the halo2 EXP-4b amplifier convention ($c \approx 1$).

8.4 Region 2.5 on a real model (EXP-2.5)

Gradient descent on the input embeddings ($\|\delta_t\| \le \varepsilon \, \|\text{emb}_t\|$) to minimize σ_k, per depth, on GPT-2. Adversarial $\sigma_{\min}$ rises with depth at every budget (slope $+0.13/+0.10/+0.08$ for $\varepsilon = 0.25/0.5/1.0$); the budget transition is visible (deep-layer massive survival 0.42–0.62 at $\varepsilon = 0.25 \to$ 0.07–0.17 at $\varepsilon = 1.0$). As stated in §6, support is qualitative: $\sigma_{\text{adv}} < m_k$ throughout, and $\ln_f$ is the weakest point.

All theory identities (decomposition, $(\star\star)$, η-decomposition, water-filling, Region-2.5 triangle bound, Gram identity) verified numerically to machine precision; the consumer scripts include synthetic self-tests separating aligned ($p \approx 2$, Gram$^+$ 1.00) from random ($p \approx 1$, Gram$^+$ 0.5) inputs before application.

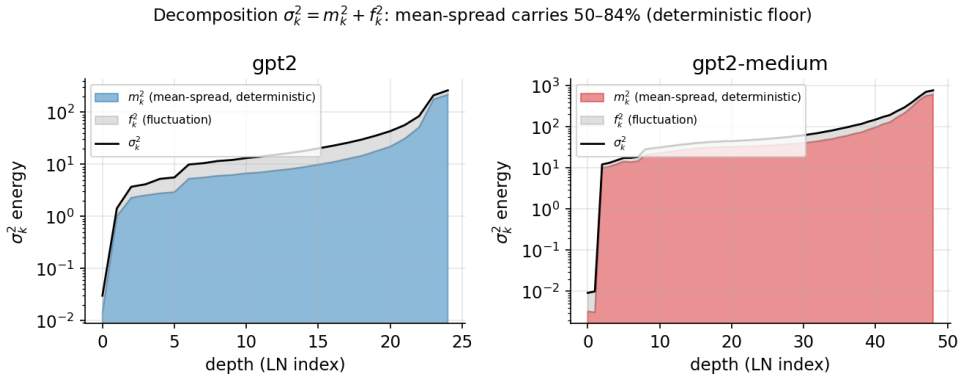


Figure 1: Decomposition $\sigma^2 = m^2 + f^2$ on GPT-2/medium: the deterministic mean-spread m_k^2 carries the bulk of the energy (≈63–83% past the first two embedding-transient layers, ≈31–36% there) and grows with depth, providing a deterministic floor.

8.5 The cost kernel and the σ-floor's load-bearing role (EXP-4a/4b/4d/5)

Kernel (EXP-4a). On EZKL/halo2 a single sigmoid lookup of r table-bits costs $\log_2(\text{prove_s}) = 1.024\,r - 16.5$ ($R^2 = 0.9976$; reconfirmed by a fresh re-measure, $1.030\,r$, $R^2 = 0.997$, EXP-B18): $K(r) \propto 2^r$, doubling per table bit. There is **no commit-dominated plateau** (halo2's monolithic table is undecomposed, the $c \approx 1$ limit), and a **build wall** appears at logrows ≈ 24–26 (three runs at logrows = 24 did not complete on 48 GB). The honest statement is required fidelity vs. build wall, not below/above a knee.

Composition (EXP-4b). The binding quantity is the per-layer required fidelity $r_\ell = \log_2 S_\ell + A$. Anchoring A to the deployed fixed-16-bit choice (zkLLM [29]/NANOZK [32], lossless on honest inputs ⇒ floored $r_\ell \approx 16$), the GPT-2 σ-floor holds every layer near 16 bits: buildable, $O(L)$. A σ-collapsed profile ($\sigma \equiv 0.1$, Regime 2) demands r_ℓ up to ≈ 85 bits: 60+ past the wall, physically unbuildable beyond depth ≈ 6, with a per-layer cost ratio $2^{a\Delta r} \approx 10^{21}$ at $L = 25$. This **supports H1** (cost is exponential in r everywhere; the floor pins r_ℓ to the deployed fidelity) and **refutes H0** (the $O(L)$ is layerwise structure, but the per-layer constant is set by σ).

Mechanism (EXP-4d). Building Jolt Atlas and reading its Shout prover (source extraction + an $r \times m$ sweep) resolves how the kernel scales. Jolt uses two orthogonal lookup pathways: a **dense** path (neural-teleport tables for sigmoid/tanh/erf) whose table-MLE is materialized, costing $O(2^r)$ with measured per-bit slope 1.0 ($c \approx 1$, matching halo2) up to a compute wall near $r \approx 28$; and a **prefix-suffix** path (ReLU/range) [20] whose cost is polynomial in the log-table-size: a logical 2^{64}-entry table proves in 82 ms, where the dense path already needs 8 s at 2^{28}. Crucially, an m-sweep is **dispositive that the table-MLE evaluation is per-proof**: at fixed $r = 24$, growing m by 1024× grows the dense-table cost gap by only 1.12× (per-lookup would demand 1024×), so the only m-scaling term is the cheap one-hot address commit $\sim d\,m$. This confirms the source-extracted Shout accounting and shows there is **no** Lasso-style $c > 1$ decomposable-table plateau.

Softmax/exp (EXP-5). The nonlinearity the floor theory is built around (softmax's exp) is implemented by Jolt not as one dense table but as a hi/lo **2-way split** with an explicit degree-3 multiplication. Sweeping the softmax element count N and the sub-table width r (36 points, all verified) cleanly separates two orthogonal cost axes. The **fidelity axis** r is a per-proof sub-table

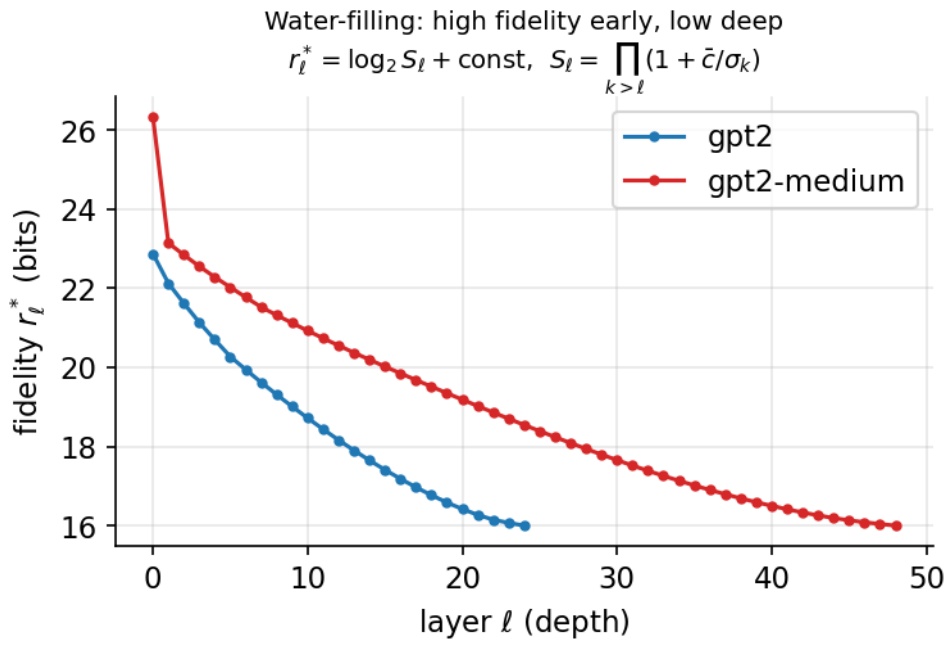


Figure 2: Water-filling allocation r_ℓ^*: high fidelity in early (small-σ, high-sensitivity) layers, low fidelity in deep (large-σ) layers.

evaluation and is essentially **free**: the 2-way split makes the table term $2 \cdot 2^{r/2}$ so small that total cost rises only 0.05 bits per bit of r, buried under a fixed ∼50 ms overhead. The **sequence axis** N carries a per-element term with slope $b \approx 0.021$ ms/element, r-independent and linear through $N = 2^{16}$ with no knee (the only wall is memory $\propto N$; proof size and rounds grow $\propto \log N$). Since 0.021 ms is ≈1.4–1.5× a single tanh lookup, **softmax is N-benign**: per element it is no more pathological than any lookup, so the σ-floor (fidelity-axis) story extends to the exp/softmax node. Honest boundary: the $\propto N$ term lies **outside** the σ-floor; it is the irreducible per-element cost of proving each exp. A full attention layer applies one softmax per query over N_{seq} keys, so it contains $\propto N_{\text{seq}}^2$ exp elements (N_{seq} = sequence length, distinct from depth L); total softmax proving is therefore $\propto N_{\text{seq}}^2$: a small constant times N_{seq}^2, not a per-element blow-up, but a genuine sequence-axis cost the floor does not govern.

The attention N_{seq}^2 limit: a degree-gap characterization (EXP-6). We asked whether the $\propto N_{\text{seq}}^2$ term could be compressed by a complementary input-geometry argument, and turn the finding into a positive characterization of why it cannot. Define the **input-geometry compression class** $\mathcal{C}$: methods that (a) peel the position-fixed sink (§7) and (b) approximate the residual exp by a degree-p polynomial, evaluating $\sum_j$ via moment tensors $T_t = \sum_j K_j^{\otimes t}$. The moment identity $\sum_j \sum_{t \le p} c_t (Q_i \cdot K_j)^t = \sum_{t \le p} c_t \langle Q_i^{\otimes t}, T_t \rangle$ is **exact** for any fixed p — an algebraic identity (verified to 10^{-16}; the low score rank $\operatorname{rank} QK^\top \le d_{\text{head}}$ is what keeps the tensors T_t small, not what makes it exact) — so $\mathcal{C}$'s only error is the polynomial approximation of exp; and $\mathcal{C}$ is content-agnostic, sidestepping the (open) soundness problem of certifying a content-selected top-k support. Two opposing degrees govern $\mathcal{C}$:

- **Affordable degree** (for sub-quadratic cost): the per-query contraction $\sum_{t \le p} \dim T_t = \sum_{t \le p} \binom{d_{\text{head}}+t-1}{t}$ must be $o(N_{\text{seq}})$, giving $p_{\text{aff}}(d_{\text{head}}, N_{\text{seq}}) = \Theta(\log_{d_{\text{head}}} N_{\text{seq}})$: measured 1 at

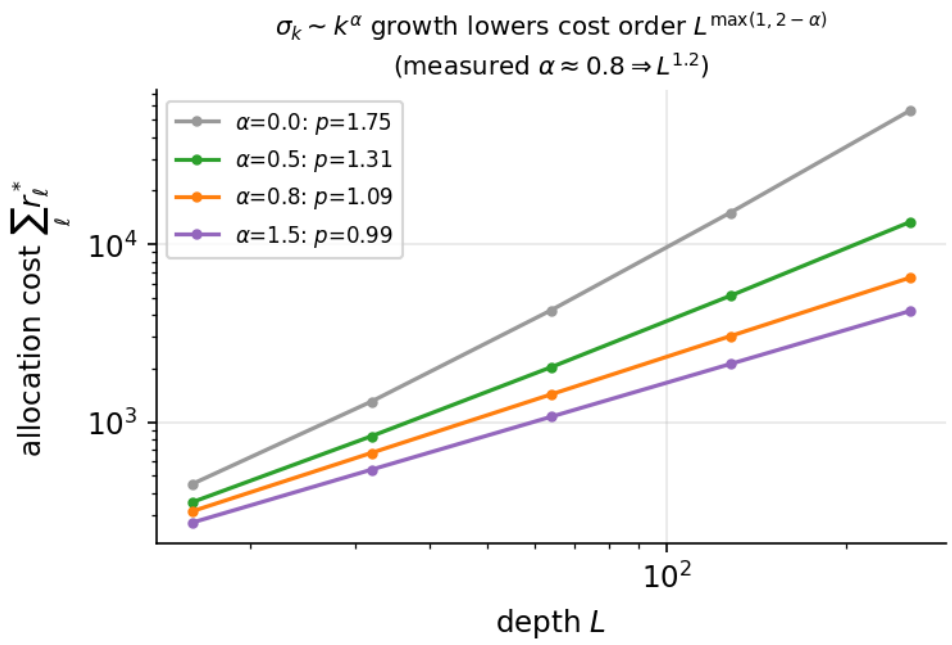


Figure 3: σ-growth lowers the cost order from L^2 (uniform floor, $\alpha = 0$) toward $L^{\max(1,2-\alpha)}$; the full-range fit $\alpha \approx 0.8$ (curve shown) gives near-linear $L^{1.2}$ (bit-as-cost model; measured $L^{0.95}/L^{1.05}$, Fig. 5). The full-range value is transient-suppressed: the bulk-window exponent is ≥ 1 on 8/10 models and the deep-half exponent ≥ 1.46 on 10/10 (EXP-A, §8.2), placing the allocation at the $\Theta(L)$ boundary of Lemma 5.

$N_{\text{seq}} = 1024$, only 3 at $N_{\text{seq}} = 2^{16}$ ($d_{\text{head}} = 64$).

- **Required degree** (for correctness): the polynomial must approximate exp to the deployment error and stay positive over the de-sinked residual range R (a sub-positive approximant makes the softmax weights non-positive and the denominator cancel). Together these force $p_{\text{req}}(R) = \Theta(R)$ — upper bound by a positive Poisson truncation, lower bound by Chebyshev growth (Appendix A) — linear in the range (measured slope ≈ 1.21, between 1 and $e/2$, $R^2 = 0.995$), and independent of N_{seq}.

Theorem 3 (attention degree gap, calibrated)**.** *On honest inputs with de-sinked residual range R, the class $\mathcal{C}$ proves attention with $o(N_{\text{seq}}^2)$ prover arithmetic — operations to build the moment sketches and contract every query, which is the class's entire cost — only if $p_{\text{aff}}(d_{head}, N_{\text{seq}}) \geq p_{\text{req}}(R)$, i.e. only in the regime $R = O(\log_{d_{head}} N_{\text{seq}})$. On real GPT-2 the de-sinked range $R_{\text{full}} \approx 9.83$ ($R_{\text{eff}} \approx 5.6$) forces $p_{\text{req}} \approx 10$–$14$, while $d_{head} = 64$ gives $p_{\text{aff}} \leq 3$ even at $N_{\text{seq}} = 2^{16}$: the condition fails by 10–14× at $N_{\text{seq}} = 1024$, still 3–5× at $N_{\text{seq}} = 2^{16}$. Hence $\mathcal{C}$ cannot compress honest-input attention below N_{seq}^2; the quadratic is tight against $\mathcal{C}$.* (Proof in Appendix A.)

The gap is worst where compression matters most: p_{req} is fixed by residual geometry (N_{seq}-independent) while p_{aff} grows only as $\log N_{\text{seq}}$, so longer context widens the relevance without closing the gap (closing it needs $N_{\text{seq}} \gtrsim \binom{78}{14} \approx 10^{15}$, the threshold N_{seq}^* below). This reproduces EXP-6's measured wall from first principles: the forced degree $p^* \approx 10$–14 is exactly the [positivity 10, accuracy 14] bracket at R_{full} (substituted perplexity $\geq +408\%$ for $p \leq 4$; raw `nan` from denominator cancellation, 41% of fitted exp-values negative at $p = 2$). The two ingredients

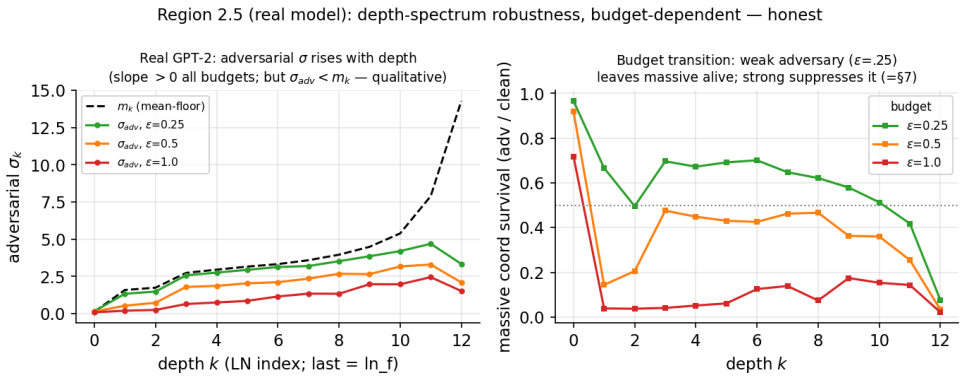


Figure 4: Region 2.5 (real GPT-2, EXP-2.5): adversarial σ_k rises with depth at all budgets but stays below the mean-floor m_k (qualitative); massive-coordinate survival shows the budget transition; $\ln_f$ is the weakest point.

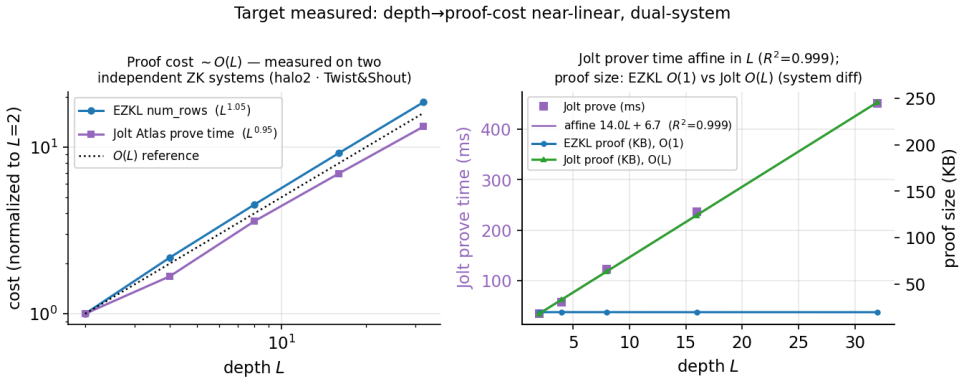


Figure 5: Measured (EXP-2): depth→proof-cost $\approx O(L)$ on two independent ZK systems (EZKL circuit work $L^{1.05}$; Jolt Atlas prover time $L^{0.95}$, affine $R^2 = 0.999$).

survive: sink mass 0.67 (across-input CV 0.02, confirming §7) and the exact moment identity. But the low-degree compression they would support does not survive the real-model residual width.

Unconditional core and robustness. The implication is itself measurement-free: $p_{\text{req}}/p_{\text{aff}} = \Theta(R/\log_{d_{\text{head}}} N_{\text{seq}})$, so $\mathcal{C}$ compresses iff $R = O(\log_{d_{\text{head}}} N_{\text{seq}})$. Only the violation is empirical, and it splits into a proven and a calibrated tier: the proven lower bound of Lemma 7 already forces $p_{\text{req}} > p_{\text{aff}}$ at $N_{\text{seq}} = 1024$ for every $R > 4.9$ — cleared several times over by the measured $R \geq 8.6$ (EXP-7 and EXP-B17, seven models) — while at $N_{\text{seq}} = 2^{16}$ the proven threshold is $R > 12.0$, and the 3–5× gap quoted there rides the calibrated slope ($p_{\text{req}} \approx 1.21R + 2.5$, EXP-6). So only a degenerate near-uniform attention escapes — range below an N_{seq}-dependent threshold (4.9 proven at $N_{\text{seq}} = 1024$; larger N_{seq} needs the calibrated slope) — and §8.5 bounds R away from zero structurally, via attention peakedness.

From a class-limited to a content-agnostic bound: a rank gap. Theorem 3 bounds the polynomial-moment class $\mathcal{C}$; we lift it to every content-agnostic method: any fixed, query-independent per-key feature map. Such a method computes $\widehat{\text{num}}_i = \alpha(q_i)^\top \Sigma$ from a shared sketch $\Sigma = \sum_j \beta(k_j) v_j^\top$, so its effective kernel $\widehat{K}_{ij} = \alpha(q_i)^\top \beta(k_j)$ has **rank** $\leq m$ by construction (cost $O(N_{\text{seq}}\, m)$; sub-quadratic iff $m = o(N_{\text{seq}})$). Because β is fixed (it sees neither the runtime values V

nor the global K), a second-moment identity (Lemma 8) forces it to approximate the kernel function, not just one instance: for value vectors that are zero-mean, uncorrelated across keys, and independent of the score geometry, $\mathbb{E}_V\|\widehat{K}V - KV\|_F^2/\mathbb{E}_V\|KV\|_F^2 = \|\widehat{K} - K\|_F^2/\|K\|_F^2$ (verified: the output error matches the kernel error to three digits). Hence δ-accuracy needs $\|\widehat{K}-K\|_F \le \delta\|K\|_F$, and by Eckart–Young–Mirsky $m \ge \mathrm{rank}_\delta(K)$. In the spherical model (q, k on a common sphere; real scores' norm spread is folded into the measured range R, and the numerical anchors below are the model's reality check), $e^{\langle q,k\rangle}$ is a dot-product kernel whose spectral decomposition is the spherical-harmonic (Funk–Hecke) one, the degree-t eigenspace having multiplicity $\binom{d_{\mathrm{head}}+t-1}{t} - \binom{d_{\mathrm{head}}+t-3}{t-2}$; so the polynomial features of $\mathcal{C}$ are optimal among separable maps there, and $\mathrm{rank}_\delta(K) = \Theta\big(\binom{d_{\mathrm{head}}+p_{\mathrm{req}}}{p_{\mathrm{req}}}\big)$ with the same $p_{\mathrm{req}} = \Theta(R)$ (an L^2 energy-rank threshold and $\mathcal{C}$'s L^∞ approximation degree: distinct quantities, argued to agree at the Θ level by a matched-scaling heuristic — both are set by the kernel's coefficient decay across R, Appendix A; verified: the numerical rank lands exactly on $(p+1)^2$ at $d_{\mathrm{head}} = 3$).

Corollary 2 (content-agnostic rank gap)**.** *No content-agnostic method proves honest-input attention in $o(N_{\mathrm{seq}}^2)$: δ-accuracy under the value model forces feature dimension $m \ge \mathrm{rank}_\delta(K)$, and in the spherical-harmonic model with the matched-scaling identification (Appendix A) this is $\Theta(F_\delta)$, $F_\delta = \binom{d_{head}+p_{\mathrm{req}}}{p_{\mathrm{req}}}$, which on GPT-2 is $\approx 9.7 \times 10^{10}$ (R_{eff}) to 1.0×10^{15} (R_{full}), vastly exceeding any practical N_{seq}. Measured directly — with no spherical model — $\mathrm{rank}_\delta(K) = 0.7$–$0.9\,N_{\mathrm{seq}} = \Theta(N_{\mathrm{seq}})$ (non-saturating through $N_{\mathrm{seq}} = 2048$), which already forces $m = \Omega(N_{\mathrm{seq}})$ and reproduces the degree gap of Theorem 3 ($\approx 13\times$ at $N_{\mathrm{seq}} = 1024$). Random-feature maps (Performer) are included: they realize the same harmonic spectrum in expectation with added variance (measured relative error 1.9–12 even at $m = 4N_{\mathrm{seq}}$).* (Proof in Appendix A.)

A finite-N_{seq} threshold. Because F_δ depends only on (R, d_{head}) and not on N_{seq}, the corollary is sharp as a threshold: content-agnostic compression is impossible for every $N_{\mathrm{seq}} < N_{\mathrm{seq}}^* := F_\delta(R, d_{\mathrm{head}})$, with $N_{\mathrm{seq}}^* \ge 1.3 \times 10^8$ even at a mild $R = 3$ (at the *calibrated* $p_{\mathrm{req}}(3) = 6$, the proven bound being weaker there; $d_{\mathrm{head}} = 64$) and $\approx 10^{15}$ at R_{full}, covering every practical context length with no asymptotic-in-N_{seq} argument.

This isolates the two genuine escapes on the count axis: **content selection** (top-k) yields a sparse (not low-rank) kernel (full numerical rank at 6% density), evading the rank bound but incurring a prover-work obligation: certifying that the $N_{\mathrm{seq}} - k$ omitted scores per query are not larger touches all keys, and because every query reorders the scores it cannot be amortized across queries (measured top-k overlap and rank-correlation across queries are at chance), so the prover does $\Omega(N_{\mathrm{seq}})$ work per query, i.e. $\Omega(N_{\mathrm{seq}}^2)$ total. For exact soundness this is unconditional in the score-comparison model: an uncompared key can be set adversarially to dominate the softmax, so each query's certificate must involve a comparison against every omitted key — $\Omega(N_{\mathrm{seq}})$ per query and, the supports being query-specific, $\Omega(N_{\mathrm{seq}}^2)$ certified comparisons in total, hence that much prover work in any system that materializes them. As a time bound in richer models (where the low-dimensional score structure could in principle be batched) the quadratic is inherited conditionally, via the orthogonal-vectors/set-disjointness hardness of exact max-inner-product in high dimension — the same barrier as below. SNARK succinctness shrinks the proof and verifier, not the prover's work. The only escape is approximate, economically-sound selection with fixed-dimension preprocessing (LSH/ANN), the same probabilistic-soundness relaxation as Fisher selection (§11), whose unconditional exclusion would require fixed-dimension nearest-neighbor lower bounds and is left open. **Model change** (linear/local attention) instead alters the function itself. Neither compresses honest standard softmax.

A fine-grained complexity complement: the barrier the prover inherits. Our degree/rank gap is class-relative (all content-agnostic methods) but unconditional. A complementary, universal barrier is inherited from the fine-grained complexity of attention computation. Alman and Song [2] prove a sharp transition at entry bound $B = \Theta(\sqrt{\log N_{\text{seq}}})$ (in the $d_{\text{head}} = O(\log N_{\text{seq}})$ regime): below it, attention admits an $N_{\text{seq}}^{1+o(1)}$-time $1/\text{poly}(N_{\text{seq}})$-additive approximation (by the polynomial method, i.e. precisely our class $\mathcal{C}$), and at it, no truly subquadratic algorithm exists unless SETH fails; the transition persists at any softmax temperature [13]. The inheritance is at the completeness stage: an honest prover must at minimum compute the attention output it proves, so any sound-and-complete proof system for standard attention on large-entry inputs has prover time $N_{\text{seq}}^{2-o(1)}$ (SETH), whatever its cryptography. This is consistent with, and independent of, our bound: their entry bound B and our de-sinked range R are the same qualitative dial (bounded scores $\Rightarrow$ low required degree), their algorithmic regime is exactly the $R = O(\log_{d_{\text{head}}} N_{\text{seq}})$ compressible branch of our unconditional core, and our measurements (EXP-7, EXP-B17: $R \geq 8.6$ across seven models, gated soft-capped Gemma-2 and Llama-3 included) land the deployed models on the hard side. The bounded-entry assumption "rarely holds in practice" is likewise the stated motivation of follow-up work [1], whose sparse-plus-dense support-basis decomposition is content-adaptive and therefore outside the class our Corollary 2 bounds (its soundness cost as a proving route is open). The useful structural fact for zkML is that the threshold is checkable: an $O(N_{\text{seq}} d_{\text{head}})$ entry-range certificate on Q, K (which bounds every score) decides the regime, so a sound protocol can certify entry bounds first and route to a certified sub-quadratic approximation only when the input is provably on the easy side; on the hard side, $\Omega(N_{\text{seq}}^2)$ honest proving is not a design failure but an inherited law. Deployed attention provers are consistent with this: zkAttn's prover is $O(K\, N_{\text{seq}}^2)$: its K-segment lookup buys precision, not count (§11).

Generality beyond GPT-2 (EXP-7). The degree/rank gap is not a LayerNorm artifact. Measuring the de-sinked residual range R, the per-row required degree, and the exp-kernel numerical rank on four RMSNorm models (Llama-family TinyLlama-1.1B and SmolLM2-360M, Qwen2-family Qwen2.5-0.5B and Qwen2.5-Coder-1.5B, head dim 64 and 128) against a GPT-2 baseline, every RMSNorm model matches or exceeds GPT-2 in range and gap: per-row $R_{\text{full}} = 8.8$–11.4 (GPT-2 8.6)[1], per-row degree gap 12–15$\times$ (GPT-2 12$\times$); and on all five models the exp-kernel rank stays a large constant fraction, $\text{rank}_\delta = 0.27$–$0.56\, N_{\text{seq}} = \Theta(N_{\text{seq}})$ (GPT-2 itself at the top, 0.56). Sink mass 0.65–0.75 (GPT-2 0.62) shows the sink-peel premise transfers as well. So the gap is robust across normalization (LayerNorm/RMSNorm), family (GPT-2/Llama/Qwen), scale (360M–1.5B), and head dimension. Scope: score geometry at $N_{\text{seq}} = 512$ on four ungated models; the gated pair is measured in EXP-B17 (Gemma-2 soft-cap and Llama-3.2, both at ungated-level R, cap non-binding at its deployed value); not an end-to-end proof benchmark.

Why R is bounded below: peakedness. The range R need not be taken as a measured constant; it is forced by how peaked attention is. For de-sinked weights $w_j = e^{s_j}/Z$ over the N'_{seq} bulk keys, let ϕ be the smallest fraction of keys carrying half the mass. Then $w_{\max} \geq (1/2)/(\phi N'_{\text{seq}})$ and $w_{\min} \leq (1/2)/((1-\phi)N'_{\text{seq}})$, so

$$R = \ln \frac{w_{\max}}{w_{\min}} \;\geq\; \ln \frac{1-\phi}{\phi}.$$

[1] Both figures are per-row de-sinked R_{full}: this EXP-7 harness reports ≈8.6 for GPT-2, the same de-sinked range Theorem 3 quotes as ≈9.83 from the EXP-6 harness, the ∼1-unit offset reflecting a window/subsampling convention (both lie in the 8.5–9.8 band), not a different statistic.

Hence $R \geq R_0$ reduces to "attention is non-uniformly peaked" ($\phi < \phi_0$), a structural, N_{seq}-independent property. Already $\phi \leq 0.05$ (half the mass on $\leq 5\%$ of bulk keys) gives $R \geq 2.9$ — on the calibrated slope, a $\geq 2\times$ degree gap through $N_{\text{seq}} = 2^{16}$ (the proven bound needs the larger measured R) — while the bound alone would already deliver the measured $R \geq 8.6$ at $\phi \approx 2 \times 10^{-4}$: the argument only ever needs the far milder $\phi \leq 0.05$; only a degenerate near-uniform attention escapes. Equivalently, with the participation ratio $n_{\text{eff}} = (\sum_j w_j^2)^{-1}$ (effective number of attended keys), a three-line argument gives $R \geq \ln(N'_{\text{seq}}/n_{\text{eff}})$, so the floor reduces to attention being selective ($n_{\text{eff}} = o(N'_{\text{seq}})$), the defining property of attention; for a local head ($n_{\text{eff}} = O(1)$) this yields $R \gtrsim \ln N'_{\text{seq}}$, making the degree-gap ratio $p_{\text{req}}/p_{\text{aff}} \to \Theta(\ln d_{\text{head}})$ a constant for all N_{seq} (neither a measured R nor a finite-N_{seq} argument is then needed). Under RoPE the logit is, moreover, a band-limited function of relative position, $s(\Delta) = \sum_i r_i \cos(\Delta\theta_i - \varphi_i)$, so this selectivity is governed by each head's rotary energy $\sum_i r_i^2$ (higher energy and higher frequency give smaller n_{eff}); only a head whose rotary energy vanishes, or sits at vanishing frequencies, is positionally non-selective. Honest caveat: this anchors R in the bulk peakedness, which is distinct from the σ-floor mechanism. The latter governs the attention sink, which is peeled and so does not bind R; the binding de-sinked range comes from locality/content routing. We therefore do not claim the depth-cost floor and the attention barrier share a single cause, only that each is independently structurally forced.

Honest scope: Corollary 2 closes content-agnostic compression but is not an unconditional $\Omega(N_{\text{seq}}^2)$: content-selected top-k escapes via a sparse-support soundness argument (exact soundness: $\Omega(N_{\text{seq}}^2)$ certified score-comparisons unconditionally, $N_{\text{seq}}^{2-o(1)}$ time under SETH; only an approximate fixed-dimension escape, Fisher-style probabilistic soundness, remains open), and model changes (linear attention) escape by replacing exp (§11). The large-entry regime is additionally covered by the SETH-conditional but method-universal inherited barrier above (Alman–Song [2]), which is ours only in the transplantation: the lower bound is theirs, the observation that a sound-and-complete prover inherits it at the completeness stage (and that the gating threshold is itself soundly checkable) is the zkML-side contribution. It rests on the honest value model above (independence of values and score geometry; under correlated values, $\mathbb{E}VV^\top = \Sigma$, the identity persists in $\Sigma^{1/2}$-weighted norms and forces $m \geq \text{rank}_\delta(K\Sigma^{1/2})$ — the bound on the principal value directions, Lemma 8 remark — and exploiting alignment requires content-adaptivity, i.e. selection). Rather than calibrate R, we anchor it in peakedness (§8.5); the R measured on GPT-2 and four RMSNorm models (EXP-7) only confirms it. The σ-floor composition above likewise uses a constructed collapse profile, and absolute r_ℓ is anchored to the deployed 16-bit rather than measured (the Jacobian c_k is measured, EXP-3; the absolute r_ℓ anchor remains).

9 The Separation Principle: What Activation Structure Can and Cannot Cheapen

The σ-floor controls the fidelity r_ℓ a nonlinearity must be proven at. This section asks the complementary question: with r fixed, can the function's own structure reduce the cost of proving it; and, precisely, which term of the node cost can structure touch? Writing the Shout node cost of §5 as

$$K(f, r) = \underbrace{a(r)}_{\text{per-proof table term } (\kappa\, c\, 2^{r/c})} + \underbrace{b \cdot m}_{\text{per-lookup address term } (\kappa_{\text{addr}}\, d\, m)},$$

we give a classification of what structure buys on $a(r)$, a separation principle showing it buys nothing on $b \cdot m$ while the lookup is kept, and a measured, deployed win on the one per-lookup dial that remains (the address width d). Numbers tagged EXP-Bn are pre-registered companion measurements on the same deployed substrate as EXP-4d/5 (all rows independently verified).

A differential-algebraic classification (prover-class–relative). This taxonomy is a supporting map, not this section's result: it locates where structure can help (the per-proof term $a(r)$), and every transformer activation lands in a cheap constructible cell, so it is not load-bearing. The separation principle below is the result: structure cannot touch the dominant per-lookup term. Fix the four representation classes the substrate actually implements (§8.5, EXP-4d): a dense table MLE ($\Theta(2^r)$ per proof, build wall $r \approx 28$), a prefix–suffix decomposition [20] ($c\,2^{r/c}$), a product/tensor path (softmax's hi/lo exp split), and a table-free algebraic constraint (witness y with $P(x, y) = 0$). Which class an activation can reach is decided by its differential-algebraic structure:

Proposition 2 (structure → cost regime, prover-class–relative)**.** *Within this four-class model: an* algebraic *activation (polynomial root, e.g.* $1/\sqrt{x}$*) admits a table-free constraint,* $\Theta(\deg \cdot m)$*, no wall, r-decoupled; an* addition-theorem *activation (*$e^{a+b} = e^a e^b$*) admits the product path (per-proof small, N-benign, EXP-5); a* holonomic *activation (a linear ODE with polynomial coefficients:* erf*, hence* $\mathrm{GELU} = \frac{1}{2}x(1 + \mathrm{erf}(x/\sqrt{2}))$ *by D-finite closure)—and, more generally, a* differentially-algebraic *activation satisfying a polynomial ODE (*tanh*, sigmoid, SiLU, each obeying a first-order algebraic Riccati ODE, e.g.* $\tanh' = 1 - \tanh^2$*; not D-finite, but ODE-certifiable all the same)—admits an ODE-certified degree-*$p(\varepsilon)$ *polynomial evaluation, table-free and range-check-free,* $\Theta(p \cdot m)$*;* periodicity/homogeneity *(RoPE* $\sin / \cos$*, scale extraction) reduces the domain of whichever cell it feeds. Only a* differentially transcendental *function (Hölder's theorem gives* Γ *as the anchor [15]) fits none of the constructible cells and is forced to the dense cell (the forcing side is made precise below: a rank invariant unifies the four classes, the existence of dense-forced functions is unconditional, and for named functions the residue reduces to an on-domain incompressibility that transformer activations provably do not meet).* (Proof in Appendix A.)

Every standard transformer activation lies in a constructible cell; the dense-forced cell is a boundary marker that transformers do not meet. The cells are empirically anchored: the algebraic $1/\sqrt{\ }$ constraint is r-invariant (no 2^r term, no wall) while its dense proxy doubles per bit (EXP-B1); the holonomic route for erf is table-free with zero range checks and $\varepsilon(p)$ certified by the ODE (EXP-B2); the addition-theorem cell is not ours to construct: the deployed softmax already uses it (§8.5). **The honest operating point:** at deployed fidelity these constructions sit within an $O(1)$ factor of the dense table, on both sides of unity: native $1/\sqrt{\ }$ is 1.28× more expensive (its two range checks carry 89% of the cost), flipping to 0.81–0.96× once the redundant check is collapsed $2 \to 1$ (EXP-B3); holonomic erf is ∼0.56× (EXP-B2), and 0.27–0.33× with the odd-symmetry rearrangement, halving its commitment count $19 \to 11$ (EXP-B5). The genuine payoff is not the constant but **fidelity–table decoupling**: past the build wall ($r > 28$, where dense erf precision is in any case capped at $\varepsilon \sim 4 \times 10^{-3}$ by the teleport scale) the algebraic/holonomic cells are the only feasible route, ∼126× cheaper near the wall, exactly the regime the σ-collapse of §8.5 would force.

Why the classification is a theorem and not a list: cost is the table's bipartition rank. The four prover classes are points on one scale. Split the r address bits into a prefix $u \in \{0,1\}^a$ and suffix $v \in \{0,1\}^{r-a}$ and rearrange the value table into the $2^a \times 2^{r-a}$ matrix $M_f[u, v] = T_f(u \,\|\, v)$. A classical lemma (the multilinear Lagrange bases on each side are independent) identifies the least number of terms in any separated representation $\tilde{f}(x_A, x_B) = \sum_{i \le s} P_i(x_A) Q_i(x_B)$ with $\mathrm{rank}(M_f)$; the ranks over the successive prefix cuts are the tensor-train ranks, and for a given cut $\mathrm{rank}(M_f)$ is exactly the number of committed polynomials a structured prover materializes.

Proposition 3 (cost = rank, model-relative)**.** *In the four-class model, the per-proof table cost of one node is* $\tilde{\Theta}(\rho \cdot 2^{r/2})$ *for a balanced two-way cut, where* $\rho = \mathrm{rank}(M_f)$*; at fidelity* ε *the operative*

quantity is the ε-approximate rank $\rho_\varepsilon = \#\{\sigma_i/\sigma_1 > \varepsilon\}$. Dense is the full-rank extreme $\rho = 2^{r/2}$; the structured classes are the small-ρ points. (Proof in Appendix A.)

Structure provably sits low on this scale, because a fixed-point value is additive across any bit cut ($\mathrm{val}(u \,\|\, v) = g(u)+h(v)$): an addition theorem gives an outer product, $\rho = 1$ (checked: the r=16 exp table has $\sigma_2/\sigma_1 = 1.6 \times 10^{-15}$; this is the deployed hi/lo split); a degree-δ polynomial factors through $\{g^0, \ldots, g^\delta\}$, $\rho \le \delta+1$ (checked for $\delta \le 7$); a holonomic function inherits $\rho_\varepsilon \le p(\varepsilon)+1$ from its certified approximant (checked: erf's ε-rank is 1–3 across the fidelity sweep, the same regime as its measured Horner commitment counts p = 7–19; the intrinsic SVD rank lower-bounds, the realizable Horner degree is what the prover pays; we do not conflate them). Genericity fills the scale:

Proposition 4 (dense-forced functions exist, unconditionally)**.** *Over a large field, the bipartition matrix of a generic table has full rank $2^{\lfloor r/2 \rfloor}$ (probability 1 for a continuous entry distribution, $1 - O(1/|\mathbb{F}|)$ over a finite field, by the order of GL_n; checked at r=14: numerical rank 128, full); by Proposition 3 it needs $\rho = 2^{\lfloor r/2 \rfloor}$ committed polynomials, i.e. dense-level per-proof cost $\tilde{\Theta}(2^r)$, in every class of the model: dense is forced.* (Proof in Appendix A.)

Whether a named function is truly dense-forced is a finer question that transformers never pose: the only natural anchor is Γ, which carries no addition-theorem, holonomic, or algebraic-differential certificate (Weierstrass [33]; Hölder [15]), whereas every standard activation carries one. The precise incompressibility condition for named anchors, and the checked SVD boundary cases, are deferred to Appendix A. Honest shape of the bounds: Propositions 3–4 are restricted-model lower bounds in the lineage of circuit/communication rank bounds; a fifth primitive could move the boundary, and the rank governs the per-proof table term only (the address axis is the separation principle's, next).

The separation principle: structure cannot move the dominant term. The classification above lives entirely in $a(r)$. That is not an accident of the cells we tried:

Proposition 5 (separation, Shout-relative)**.** *In a Shout-style lookup argument the prover's per-lookup cost is a function of the access multiset $\{a_i\}_{i\in[m]}$ alone, independent of the table values $\{T[a] = f(a)\}$. Consequently two activations $f \neq g$ evaluated on the same quantized inputs incur identical per-lookup cost, and any transformation of f that keeps the lookup acts only on the per-proof term $a(r)$, where it already receives maximal credit (one $\tilde{f}$ evaluation amortized over all m lookups). It has zero leverage on the slope b.*

Proof. The committed read-address object is the one-hot polynomial family ra whose $\sim d \cdot m$ nonzeros are determined by $\{a_i\}$ alone; the values v_i never appear in it. They enter only through the read-check sumcheck $\sum_a \tilde{f}(a)\,\mathrm{ra}(i,a) = v_i$, which references $\tilde{f}$, a per-proof object (EXP-4d, dispositive m-sweep, §8.5). Empirical confirmation of the premise: two structurally opposite functions (a smooth GELU-like vs. the singular $1/\sqrt{}$) quantized on identical inputs produce identical address arrays (27,347 distinct addresses, f-independent). □

Corollary 3 ($\Omega(m)$ commitment floor)**.** *Proving m activation evaluations requires $\Omega(m)$ commitment work ($m \cdot d$ one-hot nonzeros in Shout). Function structure amortizes the shared table, never the m per-operation commitments; a sub-$\Omega(m)$ prover requires either fewer operations (a model change) or a commitment scheme that exploits access-pattern compressibility (a different substrate).* (Proof in Appendix A.)

Two pre-registered falsification attempts anchor the principle on orthogonal channels, and a third marks its boundary. *Commitment-count channel* (EXP-B3): collapsing the $1/\sqrt{}$ range checks $2 \to 1$ removes an entire one-hot family ($33 \to 17$ commitments) and 39% of the per-proof constant, yet the slope moves only to $0.90\times$. *Pure-sharing channel* (EXP-B4): batching the two range-check sumchecks into one shares rounds and transcript without removing any commitment ($33 \to 33$); b(batched)/b(independent) $= 0.993$, with only the per-proof constant ($0.976\times$) and proof size ($0.720\times$) improving; mechanically, each round still folds every instance's length-m polynomial before combining messages, so the $\propto m$ work is untouched. The same experiment shows the backend is entropy-insensitive: driving the address multiset degenerate (distinct addresses $11{,}499 \to 8$, a $> 1400\times$ collapse) moves b by only $\sim$6%; the information-theoretic address headroom (which we measure at 10–27% for elementwise activations; only attention access patterns are strongly compressible, effective keys $5.8 \ll 512$, which is precisely the count-axis frontier of Theorem 3) is a floor no one-hot scheme realizes. *Substitution channel* (EXP-B5): replacing the lookup itself by the holonomic Horner circuit (leaving Shout, not retabulating) does move the constant, b(poly)/b(dense) $= 0.40\times$, while the order $\Theta(m)$ stands (Corollary 3 is about the order, not the constant). Honest scope: the principle is architecture-relative (Shout one-hot; Twist's read–write accounting unanalyzed), and the EXP-B4 slope ratio is a point estimate ($n = 4$ sizes; the formal CI is wide) supported by the mechanism, leave-one-out stability, and an independent EXP-B3 anchor rather than by a single confidence interval.

Cost-model inversion: the same structure is priced differently under FHE. The ingredients of Proposition 2 are individually known in private inference under fully homomorphic encryption: polynomial replacement of activations is the field's backbone (layerwise-optimized degrees [18], composite minimax sign chains whose reported state-of-the-art replaces one ReLU by compositions of total degree in the thousands), softmax is reformulated through its moment generating function [21], and Gaussian-kernel attention with LoRA makes whole architectures HE-friendly [23]. We claim none of these compressions as novel. What is new is the pricing: under FHE the cost functional is multiplicative depth $\approx \log_2$(degree), so function structure (degree, parity, kernel form) is the cost; structure has, so to speak, full leverage. Under a Shout lookup the dominant term is the per-lookup address commitment, and Proposition 5 says structure has *zero* leverage there; its entire jurisdiction is the per-proof table term, which the σ-floor has already bounded. The leverage of algebraic structure on proving cost is thus **cost-model-relative, and it inverts** between the two neighboring privacy-preserving paradigms: what is everything under a depth metric is (at the operating point) nothing under an address metric. The classification plus the separation principle make this inversion precise, and they transfer: for any backend, read off which term the structure can reach.

The remaining per-lookup dial, and a deployed win. The separation principle leaves exactly one function-adjacent dial on the per-lookup side: the address width $d = \lceil \text{bits} / \log_2 k_{\text{chunk}} \rceil$, not the table's values but the committed range of its address space, which must cover the values the computation actually reaches (the address-side analogue of $r_\ell = \log_2 S_\ell + A$: fidelity follows the reachable range). Auditing the deployed softmax node through this lens found a 22-bit mismatch: of the five one-hot families a softmax node commits (16 polynomials at scale 2^8, each of length $\propto N_{\text{seq}}^2$), the saturation-gap family `SatDiff` was range-checked at a global 32 bits, while the honest saturation gap measures 833 (10 bits) across the shipped models; the width existed solely to cover an attention-mask sentinel ($-\infty$ quantized to $\approx -2^{30}$), an encoding artifact, not a value the function produces. Re-deriving the sentinel one float unit above the exp round-to-zero score bound

$C(\nu) = \lceil(\nu{+}1)\ln 2\rceil{+}1$ (log-scale $\nu = \log_2(\text{scale})$); the resulting sentinel value is -2048 and the range-check width $\nu + 4$ bits at ν=8, making the check scale-aware: d drops $8 \to 3$, removing **5 of the 16 committed one-hot polynomials per softmax node on the $\propto N_{\text{seq}}^2$ axis**. Measured on the deployed prover (4 models, single-session A/B, medians of K=3): total one-hot commitments 121 → 116 / 209 → 199 / 433 → 413 / 1235 → 1175 (all four exactly -5/node; GPT-2 -5×12), prover time -6.0% / -4.9% / -2.6% / -4.6%, all proofs verifying, all 12 GPT-2 softmax output tensors bit-identical to the old sentinel, and masked coverage exercised on real causal-mask workloads (whole-model GPT-2: max masked gap 429, 9 bits, $\ll 2^{12}$; a second transformer export sits inside the envelope with 7.3× headroom). The narrowing is soundness-preserving: the degree-3 complementary-slackness sumcheck that pins the unique decomposition never references the range-check upper bound (the check load-bears only non-negativity/index-validity), and the completeness boundary is exact; across 12 prover runs in one build, verification passed iff the witness fit the width (4095/4096 accept/reject at ν=8), with graceful rejections. The change is upstreamed to the deployed system as a public pull request.[2] The reading we defend: this is the separation principle's positive contrapositive; the win lives on the address side precisely because it corrects the encoded range, which is the one per-lookup quantity function values do not determine; and it is a constant-factor win, exactly the class Corollary 3 permits.

Scope of validity. The classification also delimits this paper's own theory. The σ-floor cost story (§5) operates in the dense-feasible cells, r below the wall, where the address commit dominates and allocation is the game. The algebraic/holonomic constructions cover the region just past that boundary (high fidelity, wall-bound), where they are the only route. Differentially transcendental functions lie outside both, and outside transformers. In rank terms the two legs are orthogonal coordinates of one picture: the σ-floor presses the fidelity axis (ε stays loose on honest input), which keeps even a borderline finite-smoothness cell below its dense crossover, while the classification sorts which f densifies at which ε along the structure axis. The classification is thus the boundary of the σ-floor's scope plus a construction for the strip beyond it, not an engineering speedup claim; at the operating point the honest summary remains: **structure buys feasibility and constants, the floor buys the allocation, and nothing on the function axis moves the $\Theta(m)$ address term.**

10 Assumption Map (honest scope)

- **Unconditional theorems:** Lemma 1; the decomposition (1); Proposition 1 (given $\theta^* < 1$, $\nu_{\min} > 0$, for $k \geq 1$); the $\eta \leq \max(\theta, \varphi)$ bound (Remark 1; where that maximum is ≥ 0 — else $\eta \leq 0$, and "both channels $< 1 \Rightarrow \eta < 1$" holds unconditionally); water-filling KKT (given convex, differentiable, strictly increasing $\tilde{K}$ and a nontrivial budget $E^* < \sum_\ell S_\ell C$ — both kernels used qualify: the Shout per-proof table term $c\,2^{r/c}$, dense $c = 1$, and bit-as-cost r).
- **Conditional theorems:** Theorem 1 (given the alignment hypotheses (H1)–(H2)); Theorem 2 (given growth exponent / backend; $\alpha \geq 1$ measured cross-model, EXP-A, and $\alpha > 1$ not needed — Lemma 5); Region 2.5 (given input-robust massive activations); Corollary 1 (given linear σ-growth — measured in the bulk/deep windows, EXP-A — with $\beta = \bar{c}/a < 1$ assumed).

[2] `https://github.com/ICME-Lab/jolt-atlas/pull/258` (with the measurement record; merged 2026-07-07). The merged form widens the margin to ν+8 ($d\,8 \to 4$, -4 polynomials per node); all measurements reported here are at the as-submitted ν+4 form ($d\,8 \to 3$, -5 per node), commit-pinned in the measurement record, and remain reproducible against that baseline.

- **Measured facts (GPT-2/medium; cross-model where noted):** σ-growth, $\eta^* < 1$, water-filling profile; Theorem 1 directly ($\theta_k < 0$ 24/24, $p \approx 2$, Gram-positivity 1.00, §8.2); growth exponent at or above the linear boundary cross-model (EXP-A, ten models incl. two gated: deep-half $\alpha \geq 1.46$ on 10/10, bulk $\alpha \geq 0.9$ on 8/10, §8.2); depth→proof-cost $\approx O(L)$ on two ZK systems (§8.3); lookup-dominance (bit→cost flat over the measured range).
- **Structurally induced (§7), not unconditional:** massive-activation emergence & input-robustness: a training incentive; fails for sink-free architectures. This sink-free contingency is measured (EXP-B19, §7): the floor's existence and the $O(L)$ core survive, while the σ-inflation and Region 2.5 defense do not.
- **Measured but budget-dependent:** Region 2.5 (§8.4): adversarial $\sigma_{\min}$ rises with depth at all budgets, but $\sigma_{\text{adv}} < m_k$ (qualitative) and massive activations are suppressible at moderate budget ($\varepsilon \geq 0.5$).
- **Measured (EXP-4a/4b, §8.5):** the cost kernel $K(r) \propto 2^r$ on halo2 (no commit plateau; build wall logrows $\approx$ 24–26) and the σ-floor's load-bearing role (floored $r_\ell \approx 16$ bits, buildable; σ-collapse ≈ 85 bits, unbuildable past depth ≈ 6).
- **Measured + source-extracted (EXP-4d, §8.5):** Jolt Atlas's Shout cost accounting: per-proof table-MLE evaluation + an r-independent per-lookup one-hot address commit ($\sim d\,m$), no grand-product tax; per-proof confirmed dispositively by an m-sweep (cost gap grows 1.12× over 1024× m).
- **Measured (EXP-5, §8.5):** the softmax/exp node is N**-benign**: fidelity axis essentially free; per-element sequence cost $b \approx 0.021$ ms (≈1.4–1.5× a tanh lookup), r-independent and linear in N. The $\propto N$ term is outside the σ-floor; summed over an attention layer it is a small-constant $\propto N_{\text{seq}}^2$ sequence-axis cost.
- **Characterized limit (EXP-6 / Theorem 3 / Corollary 2, §8.5):** the attention N_{seq}^2 is **tight against all content-agnostic compression** (any fixed per-key feature map) by a degree/rank gap: required degree $p_{\text{req}} \propto R$ ($\approx$ 10–14 on GPT-2, reproducing EXP-6's wall) vs. affordable $p_{\text{aff}} \propto \log N_{\text{seq}}$ (≤ 3), a 10–14× gap at $N_{\text{seq}} = 1024$ (3–5× at 2^{16}); and feature dimension $m \geq \text{rank}_\delta(K)$ (Eckart–Young), $\Theta(F_\delta)$ in the spherical-harmonic model with $F_\delta = \binom{d_{\text{head}}+p_{\text{req}}}{p_{\text{req}}} \approx 10^{11}$–$10^{15} \gg N_{\text{seq}}$ (polynomial features optimal by the dot-product/spherical-harmonic structure, so random features are covered). The premises (sink mass 0.67, exact moment identity, second-moment value model) survive; the compression does not. This is a lower bound against content-agnostic methods, not an unconditional $\Omega(N_{\text{seq}}^2)$: content selection (top-k, sparse) and model change (linear) escape (§11).
- **Measured (EXP-7, §8.5):** the degree/rank gap **generalizes beyond GPT-2**: four RMSNorm models (Llama/Qwen families, 360M–1.5B, head dim 64/128) match or exceed the GPT-2 baseline in residual range ($R_{\text{full}} = 8.8$–11.4) and per-row degree gap (12–15×), with the exp-kernel rank a large constant fraction on all five models ($\text{rank}_\delta = \Theta(N_{\text{seq}})$, GPT-2 itself at the top); score geometry at $N_{\text{seq}} = 512$; gated Gemma-2/Llama-3.2 measured in EXP-B17 (below).
- **Measured (EXP-8, selectivity):** the bound $R \geq \ln(N'_{\text{seq}}/n_{\text{eff}})$ holds on all 267,840 de-sinked rows across the five models (0 violations), and real attention is selective ($n_{\text{eff}} \approx 18$–$22 \ll N'_{\text{seq}} = 256$). Honestly the bound is up to ∼5× loose (measured $R \approx 8$–11 vs. floor 1.9–2.3): selectivity alone places a positive structural floor on R; on the calibrated slope that floor corresponds to

a $\geq 2\times$ degree gap through $N_{\text{seq}} \approx 2^{12}$ (the proven bound forces no gap at $R \approx 2$), while the strong 12–15× gap still needs the measured R.

- **Measured (EXP-9, σ-floor on RMSNorm):** the floor ports to RMSNorm: RMS $\geq \sigma$ on every layer of all five models (25/25, . . . , 65/65) with the identity $\text{RMS}^2 = \sigma^2 + \text{mean}^2$ exact (err $\sim 10^{-6}$), hence $S_\ell(\text{RMSNorm}) \leq S_\ell(\text{LN})$; the RMS-floor premise (residual-norm growth ×80–449) holds. The gated variants, untested here, are covered cross-model by EXP-A (§8.2). Caveats are isomorphic to the GPT-2 σ-floor (worst-case 1/RMS confined to isolated early/sink tokens; $\theta < 0$ weaker, i.e. non-load-bearing, since the RMS-floor rests on norm growth).
- **Architecture-relative theorem + dual measured anchors (§9):** the separation principle (Proposition 5) is proved against the Shout construction (not absolute; Twist's read–write accounting unanalyzed) and anchored on two falsification channels: commitment-count (EXP-B3, slope 0.90×) and pure sharing (EXP-B4, slope 0.993×, commitments 33=33, entropy-insensitive under a 1437× address collapse). The EXP-B4 ratio is a point estimate ($n = 4$; formal CI wide) supported by mechanism, leave-one-out stability, and the EXP-B3 anchor. The substitution channel (EXP-B5, 0.40×) marks the boundary: leaving the lookup moves the constant, never the $\Theta(m)$ order.
- **Prover-class–relative classification + rank lower bound (§9):** the structure→regime map (Proposition 2) and the cost=rank invariant (Proposition 3) are relative to the four implemented representation classes (restricted-model bounds; a fifth primitive could move the boundary, the prefix–suffix precedent). Within that model: dense-forced functions exist unconditionally (Proposition 4, generic tables); for named functions dense-forcing is the two-gate criterion (certificate escape $\wedge$ on-domain incompressibility), with gate (I) proved for Γ and gate (II) an open frontier that transcendence alone does not close ($\Gamma|_{[1,5]}$ ε-rank 3, checked), and that the transformer activation set provably never meets. Measured cells: algebraic $1/\sqrt{}$ (r-invariant, no wall; operating point 1.28× → 0.81–0.96× after range-check collapse), holonomic erf (table-free, rc = 0, 0.56×; odd symmetry 0.27–0.33×; ε-rank 1–3 vs. Horner $p = 7$–19 cross-anchor).
- **Measured, deployed, upstreamed (§9):** the SatDiff address-width win: honest saturation gap 833 (10 bits) vs. a 32-bit check provisioned for a mask sentinel; scale-aware narrowing gives $d\,8 \to 3$, −5 one-hot polynomials per softmax node, prover −2.6–−6.0% (4 models, A/B, verify PASS, bit-identical outputs, exact accept/reject boundary); public PR to the deployed system. Model-dependence probed on five additional in-tree exports (EXP-B15): the −5/node drop is width-static and model-invariant — reproduced exactly on the three of the five that admit end-to-end proving (the other two hit a pre-existing upstream Cube-operand panic, tracer-level measurements only), i.e. on seven models total with the four PR models; the saturation gap is set by the mask convention (none 0, replace(−10) 834, additive 429–1453), always within the scale-aware width ($\geq 4.9\times$ headroom in the sweep; 2.82× worst-case, GPT-2).
- **Inherited, SETH-conditional (§8.5):** at entry bound $B = \Theta(\sqrt{\log N_{\text{seq}}})$ no truly subquadratic attention approximation exists (Alman–Song [2]; any temperature [13]); a sound-and-complete prover inherits the barrier at the completeness stage, so large-entry honest proving is $N_{\text{seq}}^{2-o(1)}$ universally (not just content-agnostically). The threshold is soundly checkable ($O(N_{\text{seq}} d_{\text{head}})$ entry-range certificate on Q, K); measured R (EXP-7) places deployed models on the hard side.
- **Measured (EXP-3):** absolute c_k (Jacobian): per-token diag convention, four models across two normalization families: LN medians 12/19 (GPT-2-small/medium, up to 274), RMSNorm medians 0.3/1.8 (TinyLlama-1.1B/Qwen2.5-0.5B); depth-increasing (corr +0.51–+0.76, 4/4 pos-

itive); sub-1 block counts 0/1/7/38 (fully model-dependent); seq-opnorm convention up to 640/133. Not a universal $O(1)$ constant.

- **Measured (EXP-B16):** full attention-block cost end-to-end on the deployed prover (attention-only export, $N_{\text{seq}} \in [8, 128]$, K=3): prove-time log-log slope rises monotonically 0.30 $\rightarrow$ 1.37 (large-N_{seq} OLS 1.16) toward the $\propto N_{\text{seq}}^2$ commitment-length axis, while the committed one-hot count stays N_{seq}-invariant (96 polynomials, softmax 32 = 33% by count), the count-vs-length split measured end-to-end; verifier sub-linear (26 $\rightarrow$ 98 ms); the -5/node PR win reproduces unchanged at N_{seq}=64 (-4.6% prover, verify PASS).
- **Measured (EXP-B17):** gated-model attention R: Gemma-2-2b (RMSNorm, attention-logit soft-cap 50, d_{head}=256, all 26 layers) and Llama-3.2-1B (RMSNorm, d_{head}=64, all 16 layers), N_{seq}=512: per-row R_{full} = 10.3 / 10.2 (Gemma pre-cap 10.5), sink mass 0.61 / 0.73, exp-kernel $\text{rank}_\delta \approx 0.61$ / $0.59\,N_{\text{seq}}$, degree gap 14$\times$ both: R and the degree gap inside the ungated range, sink at the GPT-2 level (0.61 vs 0.62), rank at or above the ungated top (the non-compressible direction), hard side at every layer. Gemma-2's deployed soft-cap is a model-change escape valve that does not bind at its deployed value: it narrows per-row R by 1.4%, and no residual-region score crosses the cap: the single maximum grazes it at 0.995$\times$ while the observed $q_{99.9}$ sits $\sim$4.4$\times$ below, and the cap's hard ceiling $R \leq 100$ (arithmetic, from the config constant) meets the threshold scale only at $N_{\text{seq}} \geq 2^{2500}$. Level measured at a single N_{seq}=512 (as EXP-7); N_{seq}-scaling is inherited from the selectivity bound (EXP-8), not measured here.

One line: "no collapse" is an unconditional theorem; "growth ($p \approx 2$)" and "near-linear cost ($O(L)$)" are theorems plus dual-system / direct measurement resting on a structurally-induced premise; Region 2.5 is the most conditional (budget-dependent); and the attention N_{seq}^2 is a characterized limit: tight against all content-agnostic compression by a degree/rank gap (Theorem 3, Corollary 2), not merely an untried direction. Both non-GPT-2 axes are measured: the degree/rank gap (EXP-7) and the σ-floor (EXP-9) generalize from GPT-2/LayerNorm to four RMSNorm models, the degree/rank gap further to the gated pair (EXP-B17: Gemma-2 soft-cap, Llama-3.2), and the growth exponent cross-model to ten models including both gated ones, at or above the $\alpha = 1$ boundary (EXP-A). The activation axis is classified and separated (§9): structure buys feasibility and constants, never the $\Theta(m)$ address term, with one deployed, measured exception-that-proves-the-rule on the address width (the SatDiff PR); and the large-entry attention regime carries an inherited, SETH-conditional universal prover barrier.

11 Related Work

Proving backends. The system on which we measure $O(L)$ (Jolt Atlas [7]) proves ONNX tensor ops via **Shout** (Twist & Shout [25]), a read-only lookup argument using one-hot addressing. We extracted its prover-cost accounting from the source: an r-independent per-lookup one-hot address commit ($\sim d\,m$) plus an r-dependent per-proof table-MLE evaluation (dense 2^r, or prefix-suffix $c\,2^{r/c}$ [20]), with no grand-product / memory-checking tax. The earlier lookup-centric SNARKs Lasso [26] and Jolt [3] decompose tables differently (committing $3cm + c\,n^{1/c}$), but their r-dependent term has the same $c\,2^{r/c}$ shape, which is why a Lasso-framed kernel reproduces our scaling while mis-describing the mechanism that Shout does not incur. These works give the cost model; we supply the input-distribution theory that determines where on the kernel a real model sits.

zkML for transformers. zkLLM [29] introduced lookup approximations for the non-arithmetic operators (softmax, GELU, LayerNorm) under a controlled error budget; NANOZK [32] made

the proof layerwise (each layer a constant-size proof independent of width, so total cost is $O(L)$ by construction) and added Fisher-information layer prioritization. DeepProve (Lagrange) and EZKL [36] are end-to-end systems; zkLoRA [24] targets LoRA-adapter verification; ZKML [8] is an optimizing compiler, and a recent survey [22] maps the landscape.

Boundary with the layerwise line (NANOZK). The near-linear depth scaling itself is not our contribution; layerwise proving already yields it. The one real overlap is NANOZK's Fisher-information prioritization, and stating it precisely matters, because the two methods act on orthogonal axes. NANOZK ranks layers by the Fisher trace $I_\ell = \mathrm{tr}(\mathcal{F}_\ell)/|\theta_\ell|$ to choose which layers to verify (top-k layer selection), holding the lookup precision uniform at a fixed 16 bits; it never varies bits across layers. Water-filling does the converse: it verifies every layer but sets the per-layer fidelity $r_\ell^* \propto \log_2 S_\ell$. Selection and precision are different axes that compose rather than compete. What is new here is thus (i) an a-priori error→cost reduction identifying $1/\sigma$ as the sole *unbounded* data-dependent amplifier; (ii) a derivation of the allocation order and the depth-cost exponent from residual-stream geometry rather than from measured gradient statistics; and (iii) an account of why a fixed per-layer fidelity suffices as depth grows (the σ-floor keeps S_ℓ bounded), giving theoretical license to NANOZK's empirically "surprising" 16-bit, zero-perplexity-loss choice. Our threat model further explains the security boundary NANOZK states for its heuristic: Fisher selection is, in their words, "an efficiency optimization, not a security guarantee," sound only against economically-motivated adversaries and forced to fall back to full verification against a worst-case one. That is precisely our typical-vs-worst-case axis: Fisher partial verification lives in the honest-input regime (Regime 3), and the adversary that defeats it is the same σ-collapsing, sensitivity-spiking adversary of Regime 2; so the mandated fall-back is the predicted behavior at the regime boundary, not an ad-hoc patch. The same probabilistic-soundness trade reappears on the sequence axis as content-selected top-k attention (Table 2), which escapes the exact per-query $\Omega(N_{\mathrm{seq}})$ (total $\Omega(N_{\mathrm{seq}}^2)$) selection-soundness obligation only under the identical economically-motivated-adversary assumption.

Error propagation adjacent to proving. Two 2026 results triangulate our mechanism from neighboring questions and sharpen what is and is not ours. Zamir [35] shows layerwise approximate verification is non-composable in general: for any network there is a functionally equivalent one on which adversarially chosen, tolerance-passing per-layer errors steer the output arbitrarily (within a prescribed bounded range). His adversary chooses the errors themselves, a strictly stronger corruption model than our input-only adversary, so the correct reading is not that pre-LN transformers escape the counterexample, but that per-layer tolerances alone certify nothing: a whole-model propagation argument is the load-bearing step. Sections 2–4 supply exactly that argument (sensitivity calculus, σ-floor, and the threat-model split that prices the worst case separately). Independently, Basu et al. [6] reach the same contraction physics from the compression side: residual-stream growth outpaces injected weight-compression error, in a Lyapunov-style contraction analysis with machine-checked (Lean 4) per-matrix bounds, confirming the residual-growth premise the σ-floor rests on, in a setting with no prover, no fidelity allocation, and no adversary; the per-layer expansion cases reported there ($\rho_{\max} > 1$ on some architectures) match our product-form upper bound reading rather than any universal per-layer contraction claim. Two further recent results, outside zkML, corroborate the premise from the precision side: a layer-wise low-precision stability theory [4] proves a residual relaxation inequality (residual blocks attenuate depth-wise error accumulation) with a precision- and width-aware LayerNorm indicator, and a post-training quantization method [19] finds that normalization keeps the error between normalized sub-block outputs from growing be-

yond a certain depth. Both are upper-bound (or empirical attenuation) statements about forward error; our σ-floor is the complementary lower bound on σ (Proposition 1, Theorem 1) that caps the amplifier, supplying a mechanism for the depth-boundedness those works report as a symptom (Corollary 1 turns the LoaQ direction into a formal consequence of the growth-with-floor mechanism). The amplifier-and-residual physics is thus independently established; what is ours is the σ-floor decomposition and, above all, the pricing of that physics as ZK proof cost.

Attention compression: three orthogonal cost axes. The proving cost of an attention layer factors along three axes, and existing accelerators each attack a different one (Table 1). The *count* axis (N_{seq}^2 exp elements) is the one Theorem 3 addresses: on honest standard softmax it is tight against the input-geometry class. Performer [9] escapes only by replacing exp with a random-feature map $\varphi(Q) \cdot \psi(K)$ (our moment-tensor expansion is its exact deterministic counterpart, and fails at real residual widths); content-selected top-k escapes only with a still-open sparse-support soundness argument. The *fidelity/unit* axis (cost of one exp) is what our σ-floor governs and what zkAttn's digit-lookup softmax [29] cheapens; both leave the count unchanged (consistent with our N-benign EXP-5); asymptotically zkAttn's prover is $O(K\,mn) = O(K\,N_{\text{seq}}^2)$ (the full $N_{\text{seq}} \times N_{\text{seq}}$ score matrix is materialized and committed; the K-segment decomposition buys precision over bit-decomposition, not count), which is not a design flaw but the inherited large-entry law (§8.5). The *gauge* axis (weight-space symmetry) is GaugeZKP's [11] lever, composable with but distinct from the other two. Placing these on one map is itself a contribution: the count axis on honest input is the hard one, which explains why deployed accelerators target the other two.

Method	Cost axis	Lever	On honest standard softmax
This work (σ-floor)	fidelity (r)	residual σ-geometry bounds S_ℓ	governs (allocation order)
This work (EXP-5)	unit (per-exp)	node is N-benign	measured benign
This work (Thm 3)	**count (N_{seq}^2)**	**degree gap**	**tight: no compression**
zkAttn [29]	unit (per-exp)	digit-lookup softmax	cheaper unit, count unchanged
Performer [9]	count	random-feature map	covered by rank gap (Cor. 2)
top-k / sparse	count	content selection	exact $\Omega(N_{\text{seq}}^2)$ (certified comparisons); approx escape
GaugeZKP [11]	gauge (params)	weight-space symmetry	orthogonal lever

Table 1: Attention proving cost factors into three orthogonal axes; existing accelerators each attack a different one. The count axis (N_{seq}^2) on honest standard softmax is tight against the input-geometry compression class (Theorem 3), so deployed accelerators target the fidelity and gauge axes instead.

The count axis: three escape routes, three barriers. Our reduction (Corollary 2) organizes every way to break N_{seq}^2 on honest standard softmax into three mutually exclusive routes, each blocked differently (Table 2). A *content-agnostic* sketch is a rank-m kernel approximation, blocked by the **rank** gap ($m \geq F_\delta \gg N_{\text{seq}}$, polynomial features optimal so random features are covered);

content selection (top-k) is sparse rather than low-rank, evading the rank bound but blocked by a **soundness** obligation (certifying the omitted scores); a *model change* (linear/local attention) escapes only by altering the **function**. This locates Performer and moment/random-feature methods in the first route, top-k/sparse in the second, and linear attention in the third.

Escape route	Kernel structure	Barrier	Honest softmax
Content-agnostic (moments, RF, Performer)	rank-m (low rank)	**rank**: $m \geq F_\delta \gg N_{\text{seq}}$	blocked (Cor. 2)
Content-selected (top-k)	sparse, full-rank	**soundness**: certify omitted scores	exact: $\Omega(N_{\text{seq}}^2)$ certified score-comparisons (uncond.); approx=Fisher escape
Model change (linear/local)	different kernel	**function**: replaces exp	not standard softmax

Table 2: The three escape routes on the count axis and their distinct barriers (Corollary 2). Content-agnostic compression is low-rank by construction and blocked by the rank gap; the only routes that evade it abandon either soundness-for-free (selection) or the softmax function (model change).

FHE-based private inference: the same structures, an inverted cost model. The nearest neighboring paradigm prices the same function structures oppositely, which is why we are careful about what §9 does and does not claim. Polynomial replacement of activations is the backbone of FHE private inference: from composite minimax sign chains (the state of the art the layerwise line reports replaces a single ReLU by compositions of total degree in the thousands) to layerwise-optimized degrees driven by each layer's actual input distribution [18], the FHE analogue of our water-filling (degrees allocated per layer there, bits per layer here). Softmax has been reformulated through its moment generating function for HE [21], and Gaussian-kernel attention plus LoRA yields encryption-friendly LLM architectures [23]. None of the individual compressions of §9 (parity, polynomial replacement, range reduction, kernel surrogates) is novel as a compression; what is new is the pricing theorem: under a depth metric structure is the whole cost, under a Shout address metric it has zero leverage on the dominant term (Proposition 5); and the soundness dimension (a malicious prover, an accept/reject boundary) has no FHE counterpart at all, which is where the deployed SatDiff win and the checkable attention threshold live.

Massive activations and quantization. The phenomenon our mechanism rests on is well documented: massive activations [30] are a few input-agnostic, near-constant hidden units acting as implicit attention biases, related to outlier features [10, 17] and to attention sinks [34, 12]. Sink-aware quantizers (RotateKV [27], KVSink [28]) protect these coordinates as a precision burden. **Our use of the same phenomenon has the opposite sign:** the large σ produced by massive activations shrinks the LayerNorm gain $1/\sigma$, bounding sensitivity and making proving cheap. This also locates the main risk to our mechanism: Softpick [37], a sink-free rectified-softmax replacement, would remove the super-linear σ-growth and the Region-2.5 premise, though not the floor's existence (§7).

12 Conclusion

For average-case / honest-input transformer inference, ZK proof cost is polynomial (and, on lookup-based backends, near-linear $O(L)$) because the residual σ-floor (a deterministic mean-spread floor plus alignment-driven growth) keeps the fidelity allocation in the address-commit-dominated regime. We **measure** this depth→cost scaling on two independent ZK systems (EZKL and Jolt Atlas) and find $\approx O(L)$ in both, and we confirm the growth mechanism (Theorem 1, $p \approx 2$) directly on real models, with the growth exponent re-measured on ten models at or above the $\alpha = 1$ boundary, where the allocation closes with no unproven $\alpha > 1$ (EXP-A, Lemma 5). The mean floor additionally provides a partial (but budget-dependent) adversarial robustness, conditioning rather than contradicting the worst-case impossibility. We have separated unconditional theorems, conditional theorems, measured facts, and structurally-induced premises. The remaining steps are narrower: the softmax/exp node is measured and N-benign (EXP-5), the full attention block is benchmarked end-to-end (EXP-B16: prover slope rising monotonically toward the quadratic commitment-length axis, committed count N_{seq}-invariant), and its $\propto N_{\text{seq}}^2$ sequence cost lies outside the σ-floor, which we **characterize as a degree/rank gap** (Theorem 3, Corollary 2): any content-agnostic compression is a rank-m kernel approximation needing degree $\propto$ the residual range ($\approx$ 10–14 on GPT-2): measured directly, $\text{rank}_\delta = \Theta(N_{\text{seq}})$ forces $m = \Omega(N_{\text{seq}})$, and in the spherical-harmonic model (matched-scaling) $m \geq F_\delta \gg N_{\text{seq}}$; yet it can afford only $\propto \log N_{\text{seq}}$ (≤ 3); polynomial features are optimal in that model (dot-product/spherical-harmonic), so the bound covers random features and the quadratic is tight against all content-agnostic methods, the only escapes being content selection (soundness) and model change (function), and the three attention cost axes (count, fidelity, gauge) are separated. This gap is not a GPT-2 artifact: it generalizes across four RMSNorm models (Llama/Qwen, 360M–1.5B; EXP-7), and on the large-entry side it is doubled by an inherited, SETH-conditional universal barrier (Alman–Song, transplanted to the prover at the completeness stage, with a soundly checkable threshold). On the activation axis, a differential-algebraic classification and a Shout-relative separation principle (§9) close the remaining direction: function structure buys high-fidelity feasibility and $O(1)$ constants (never the $\Theta(m)$ per-lookup address term), and its leverage inverts between FHE's depth metric and the lookup-SNARK address metric; the one remaining address-side dial (committed width) we converted into a measured, deployed, soundness-preserving win (upstreamed as a public PR: -5 committed polynomials per softmax node, -2.6–-6.0% prover time on the four PR models, -0.2–-11.3% on three further in-tree exports; the invariant is the -5 per node). What is left is the absolute prover constants (gated-model attention R: measured, EXP-B17).

A Proofs

This appendix proves in full every result whose statement carries a forward pointer, and proves in place the two results the main text marks as immediate: the deterministic floor Proposition 1 (induction on ($\star\star$)) and the separation Proposition 5 (the read-address object). We reuse the notation of the section in which each result appears, and number supporting lemmas continuing the main text. Measured constants (the Jacobian norms c_k, the de-sinked attention range R, the numerical ranks) enter only as empirical inputs quoted from the theorem statements; the arguments below are the analytic scaffolding that turns those measurements into the stated scalings and are themselves measurement-free. The one place where a step is an empirical calibration rather than a derivation, the exact slope of the required attention degree, is flagged as such.

A.1 The LayerNorm amplifier (Lemma 1)

Lemma 2 (LayerNorm Jacobian)**.** *Let $x \in \mathbb{R}^d$ have coordinate mean $\mu = \frac{1}{d}\mathbf{1}^\top x$, let $P = I - \frac{1}{d}\mathbf{1}\mathbf{1}^\top$ be the mean-removal projection, and $\sigma = \|Px\|/\sqrt{d} > 0$. For $\mathrm{LN}(x) = \gamma \odot (x - \mu\mathbf{1})/\sigma$ (elementwise gain γ; the additive shift has zero Jacobian),*

$$J_{\mathrm{LN}}(x) = \tfrac{1}{\sigma}\,\mathrm{diag}(\gamma)\,M, \qquad M := I - \tfrac{1}{d}\mathbf{1}\mathbf{1}^\top - \tfrac{1}{d}\,zz^\top, \quad z := \tfrac{x-\mu\mathbf{1}}{\sigma},$$

and $\|J_{\mathrm{LN}}(x)\| \le \|\gamma\|_\infty/\sigma$, in which $1/\sigma$ is the only x-dependent factor.

Proof. Put $y = x - \mu\mathbf{1} = Px$, so $\mathbf{1}^\top y = 0$, $\sigma^2 = \frac{1}{d}\|y\|^2$, and $\mathrm{LN}(x)_i = \gamma_i y_i/\sigma$. From $\partial y_i/\partial x_j = \delta_{ij} - \frac{1}{d}$ and, using $\mathbf{1}^\top y = 0$, $\partial\sigma/\partial x_j = \frac{1}{d\sigma}\sum_i y_i(\delta_{ij} - \frac{1}{d}) = \frac{y_j}{d\sigma}$, the quotient rule gives

$$\frac{\partial}{\partial x_j}\frac{y_i}{\sigma} = \frac{1}{\sigma}\Big(\delta_{ij} - \tfrac{1}{d}\Big) - \frac{y_i}{\sigma^2}\cdot\frac{y_j}{d\sigma} = \frac{1}{\sigma}\Big(\delta_{ij} - \tfrac{1}{d} - \tfrac{z_i z_j}{d}\Big), \qquad z = \tfrac{y}{\sigma}.$$

Scaling row i by γ_i yields the displayed $J_{\mathrm{LN}} = \frac{1}{\sigma}\mathrm{diag}(\gamma)M$. Now $\frac{1}{d}\mathbf{1}\mathbf{1}^\top$ and $\frac{1}{d}zz^\top$ are the orthogonal projections onto the lines $\mathbb{R}\mathbf{1}$ and $\mathbb{R}z$ (indeed $\|\mathbf{1}\|^2 = d$ and $\|z\|^2 = \|y\|^2/\sigma^2 = d$), and they are mutually orthogonal because $\mathbf{1}^\top z = \frac{1}{\sigma}\mathbf{1}^\top y = 0$. Hence M is the orthogonal projection onto $\{\mathbf{1}, z\}^\perp$: symmetric, idempotent, spectrum $\subseteq \{0, 1\}$, so $\|M\| \le 1$ (with equality once $d \ge 3$). Therefore $\|J_{\mathrm{LN}}\| \le \frac{1}{\sigma}\|\mathrm{diag}(\gamma)\|\,\|M\| \le \|\gamma\|_\infty/\sigma$, and since γ is a frozen parameter the sole x-dependence is the scalar $1/\sigma$. □

Proof of Lemma 1. With $x_{k+1} = x_k + g_k$ and $g_k = \mathrm{Block}_k(\mathrm{LN}(x_k))$, the one-block Jacobian is $B_k = I + J_{\mathrm{Block},k}(u)\,J_{\mathrm{LN},k}$ evaluated at $u = \mathrm{LN}(x_k)$. By Lemma 2, $\|J_{\mathrm{LN},k}\| \le \|\gamma_k\|_\infty/\sigma_k$, so

$$\|B_k - I\| \le \|J_{\mathrm{Block},k}(u)\|\,\|\gamma_k\|_\infty/\sigma_k =: c_k/\sigma_k, \qquad c_k = \|J_{\mathrm{Block},k}(u)\|\,\|\gamma_k\|_\infty.$$

Lemma 2 collects every factor scaling as an inverse activation scale into the single $1/\sigma_k$; the block weights and the gain are frozen, so the only unbounded data-dependent amplifier is $1/\sigma_k$ (the block Jacobian's dependence on $u = \mathrm{LN}(x_k)$ enters c_k, which is bounded independently of the data: u ranges over the LayerNorm image $\{\gamma_k \odot z : \|z\| = \sqrt{d},\ \mathbf{1}^\top z = 0\}$ — up to the frozen additive shift — a compact set, so for a continuously differentiable block $\sup_u \|J_{\mathrm{Block},k}(u)\| < \infty$; the MLP chain bound below makes this quantitative). For an MLP block $g = W_2\,\mathrm{GELU}(W_1 u)$ the constant admits the fully a-priori bound $c_k \le \|\gamma_k\|_\infty\,\|W_2\|\,\|W_1\|\,\mathrm{Lip}(\mathrm{GELU})$ with $\mathrm{Lip}(\mathrm{GELU}) = \sup_x|\mathrm{GELU}'(x)| \approx 1.13$; the operator-norm chain is loose, which is why the deployed c_k are quoted as measured (EXP-3) rather than from this bound. □

A.2 Alignment-driven growth (Theorem 1)

Proof of Theorem 1. Since $P\mu_0 = 0$ and $\mu_{k+1} = \mu_k + \gamma_k$, additivity gives $P\mu_k = \sum_{j<k} P\gamma_j$, so

$$\langle P\mu_k, P\gamma_k\rangle = \sum_{j<k}\langle P\gamma_j, P\gamma_k\rangle = \sum_{j<k} G_{jk},$$

and because $d, m_k, \nu_k > 0$ in $\theta_k = -\langle P\mu_k, P\gamma_k\rangle/(d\,m_k\nu_k)$, we have $\theta_k \le 0 \iff \sum_{j<k} G_{jk} \ge 0$. (If $m_k = 0$, θ_k is undefined; then the cross term vanishes and $(\star\star)$ reads $m_{k+1}^2 = m_k^2 + \nu_k^2$ directly, the $\theta_k = 0$ case of the bound below — in particular the $k = 0$ start, where $P\mu_0 = 0$.) Writing $P\gamma_j = a_j u + v_j^\perp$ with u a unit vector and $u \perp v_j^\perp$ gives (using $\|u\|^2 = 1$) $G_{jk} = a_j a_k + \langle v_j^\perp, v_k^\perp\rangle$, hence

$$\sum_{j<k} G_{jk} = \sum_{j<k} a_j a_k + \sum_{j<k}\langle v_j^\perp, v_k^\perp\rangle \ \ge\ \sum_{j<k} a_j a_k - \Big|\sum_{j<k}\langle v_j^\perp, v_k^\perp\rangle\Big| \ \ge\ 0,$$

where the last inequality is exactly hypotheses (H1) ($a_j \geq 0$, so $\sum a_j a_k \geq 0$) and (H2). Thus $\theta_k \leq 0$. Substituting into ($\star\star$) and expanding, $m_{k+1}^2 = m_k^2 - 2\theta_k \nu_k m_k + \nu_k^2 \geq m_k^2 + \nu_k^2$ since $-2\theta_k \nu_k m_k \geq 0$: the mean-spread grows (strictly when $\nu_k > 0$). For the rates, with $0 < \nu_- \leq \nu_k \leq \nu_+$ (the hypothesis $\nu_k = \Theta(1)$): full alignment, $\theta_k = -1$, gives $m_{k+1} = m_k + \nu_k$, so $k\nu_- \leq m_k \leq m_0 + k\nu_+$ and $m_k^2 = \Theta(k^2)$; no alignment, $\theta_k = 0$ (orthogonal increments), gives $m_{k+1}^2 = m_k^2 + \nu_k^2$, so $m_k^2 = \Theta(k)$ (a random walk in $P\mu$). □

A.3 End-to-end cost (Theorem 2)

Lemma 3 (water-filling allocation)**.** *Let $\tilde{K} : \mathbb{R}_{\geq 0} \to \mathbb{R}$ be convex, differentiable, and strictly increasing, let $S_\ell, C > 0$, and assume $E^* < \sum_\ell S_\ell C$ (otherwise $r^* = 0$ is trivially optimal). The program $\min_{r \geq 0} \sum_\ell \tilde{K}(r_\ell)$ subject to $\sum_\ell S_\ell C\, 2^{-r_\ell} \leq E^*$ (the per-layer error is $\varepsilon_\ell = C\, 2^{-r_\ell}$) is convex, and at its optimum a single multiplier $\lambda > 0$ satisfies the stationarity $\tilde{K}'(r_\ell^*) = \lambda \ln 2 \cdot S_\ell C\, 2^{-r_\ell^*}$ at every unclipped layer. In particular $r_\ell^* = \frac{c}{c+1}\big(\log_2 S_\ell + A\big)_+$ for the Shout table term $\tilde{K}(r) = \kappa c\, 2^{r/c}$, and $r_\ell^* = \big(\log_2 S_\ell + A\big)_+$ for the bit-as-cost term $\tilde{K}(r) = r$, with A a layer-independent constant; in both cases $r_\ell^* = \Theta(\log_2 S_\ell)$, and the optimizer is unique.*

Proof. Each $\tilde{K}(r_\ell)$ is convex and each constraint summand $S_\ell C\, 2^{-r_\ell}$ is convex in r_ℓ, so the objective is convex over a convex feasible set; a minimizer exists ($\tilde{K}$ is convex and strictly increasing, hence coercive, so the search restricts to a compact sublevel set), and Slater's condition holds (large r is strictly feasible), so the KKT conditions are necessary and sufficient. Stationarity of the Lagrangian $\sum_\ell \tilde{K}(r_\ell) + \lambda\big(\sum_\ell S_\ell C\, 2^{-r_\ell} - E^*\big)$ reads $\tilde{K}'(r_\ell) - \lambda \ln 2\, S_\ell C\, 2^{-r_\ell} = 0$; monotonicity of $\tilde{K}$ and $E^* < \sum_\ell S_\ell C$ make the energy budget active at any optimum (otherwise some $r_\ell > 0$ could be reduced, decreasing the objective), so $\lambda > 0$. Substituting $\tilde{K}(r) = \kappa c\, 2^{r/c}$ gives $\kappa \ln 2\, 2^{r_\ell/c} = \lambda \ln 2\, S_\ell C\, 2^{-r_\ell}$, i.e. $2^{r_\ell(c+1)/c} = \lambda S_\ell C/\kappa$ and the stated closed form; $\tilde{K}(r) = r$ gives $1 = \lambda \ln 2\, S_\ell C\, 2^{-r_\ell}$, i.e. $2^{r_\ell} = \lambda \ln 2\, S_\ell C$. The nonnegativity clip $(\cdot)_+$ handles layers driven to $r_\ell^* = 0$. Since S_ℓ decreases with depth, r_ℓ^* is largest at small ℓ: high fidelity early, low fidelity deep. Uniqueness: suppose $r^1 \neq r^2$ are both optimal. The budget function $g(r) = \sum_\ell S_\ell C\, 2^{-r_\ell}$ has Hessian $\operatorname{diag}\big((\ln 2)^2 S_\ell C\, 2^{-r_\ell}\big) \succ 0$, so it is strictly convex, and (as above) $g = E^*$ at both optima. The midpoint $\bar{r}$ then satisfies $g(\bar{r}) < E^*$ with $\sum_\ell \tilde{K}(\bar{r}_\ell)$ no larger than the optimal value; since $g(0) = \sum_\ell S_\ell C > E^*$ forces $\bar{r} \neq 0$, some coordinate of $\bar{r}$ can be strictly reduced while keeping $g < E^*$, strictly decreasing the objective ($\tilde{K}$ strictly increasing) — contradicting optimality. □

Lemma 4 (depth-sum asymptotics)**.** *Let $\sigma_k \sim \sigma_0 k^\alpha$ with $\alpha \geq 0$, $c_k = \Theta(1)$ (bounded above and below across depth; measured $c_k \in [0.08, 274]$ cross-model, EXP-3), and $S_\ell = \prod_{k>\ell}(1 + c_k/\sigma_k)$. (The orders below are stated for $c_k = \Theta(1)$; when c_k instead grows with depth the governing quantity is $\sum_{\ell<k\leq L} c_k/\sigma_k$, so the exponent tracks the decay of c_k/σ_k rather than that of σ_k^{-1} alone.) Then $\log_2 S_\ell = \Theta\big(\sum_{\ell<k\leq L} \sigma_k^{-1}\big)$, and the fidelity-driven allocation is*

$$\sum_{\ell=1}^{L} (\log_2 S_\ell)_+ = \begin{cases} \Theta(L^{2-\alpha}) & 0 \leq \alpha < 1 \quad (\Theta(L^2) \text{ at } \alpha = 0), \\ \Theta(L) & \alpha = 1, \\ \Theta(L^{2-\alpha}) = o(L) & 1 < \alpha < 2, \\ \Theta(\log L) & \alpha = 2, \\ \Theta(1) & \alpha > 2. \end{cases}$$

Hence the water-filling optimum $r_\ell^* = (\log_2 S_\ell + A)_+$ has $\sum_\ell r_\ell^* = O\big(L^{\max(1,\, 2-\alpha)}\big)$, dropping from the no-floor L^2 at $\alpha = 0$; a fixed positive per-layer baseline $A > 0$ contributes a further $\Theta(L)$, making the allocation $\Theta(L)$ for every $\alpha \geq 1$.

Proof. Because c_k/σ_k lies in a bounded range $[0, T_{\max}]$, $\log(1+t) = \Theta(t)$ there ($\frac{\log(1+T_{\max})}{T_{\max}}\, t \le \log(1+t) \le t$), and $c_k = \Theta(1)$ (the lower bound $c_k \ge c_{\min} > 0$ makes each summand $\Theta(\sigma_k^{-1})$), so $\log_2 S_\ell = \sum_{k>\ell} \log_2(1 + c_k/\sigma_k) = \Theta\big(\sum_{\ell<k\le L} \sigma_k^{-1}\big)$. With $\sigma_k \sim \sigma_0 k^\alpha$, sum the tails by exchanging the order of summation:

$$\sum_{\ell=1}^{L} \sum_{\ell<k\le L} \sigma_k^{-1} = \sum_{k=2}^{L} (k-1)\, \sigma_k^{-1} = \Theta\Big(\sum_{k\le L} k^{1-\alpha} \Big),$$

which by integral comparison is $\Theta(L^{2-\alpha})$ for $\alpha < 2$ (in particular $\Theta(L^2)$ at $\alpha = 0$ and $\Theta(L)$ at $\alpha = 1$), $\Theta(\log L)$ at $\alpha = 2$, and $\Theta(1)$ for $\alpha > 2$. (Per ℓ, integral comparison gives the upper bounds $\sum_{\ell<k\le L} \sigma_k^{-1} = O(L^{1-\alpha} - \ell^{1-\alpha})$ for $\alpha < 1$, $O(\log(L/\ell))$ for $\alpha = 1$, and $O(\ell^{1-\alpha})$ for $\alpha > 1$; the last is not a uniform two-sided estimate near $\ell = L$ — at $\ell = L-1$ the tail is the single term $\Theta(L^{-\alpha})$ — which is why the exchange, not a per-ℓ estimate, carries the sum.) Since $r_\ell^* = (\log_2 S_\ell + A)_+$ adds at most A per unclipped layer, the total is the fidelity part plus $O(L)$, so its order is $O(L^{\max(1,2-\alpha)})$, set by the early layers, and equals $\Theta(L^{2-\alpha})$ for $\alpha \le 1$; the clip only drops deep layers with $\log_2 S_\ell + A < 0$. □

Lemma 5 (the allocation closes at the linear boundary)**.** *Assume the hypotheses of Corollary 1: $\sigma_k \ge a\,k$ for $k \ge k_0$ (a linear floor, $\alpha \ge 1$), $\sigma_k \ge \sigma_{\min} > 0$ elsewhere, $\bar{c} = \sup_k c_k$, and $\beta = \bar{c}/a < 1$. Then $S_\ell \le C_0 (L/\ell)^\beta$ with $C_0 = (1 + \bar{c}/\sigma_{\min})^{k_0}$, and for any water-filling level $A = O(1)$ (the relative-budget convention at $\alpha \ge 1$, §5):*

(i) the per-layer optimum obeys $r_\ell^* \le \beta \log_2(L/\ell) + O(1)$, which at exactly linear growth ($\sigma_k = \Theta(k)$, $c_k = \Theta(1)$) is attained up to constants at $\ell = O(1)$ (Lemma 4's two-sided estimate), so per-layer boundedness genuinely needs $\alpha > 1$;

(ii) the summed dense table term nevertheless closes:

$$\sum_{\ell=1}^{L} 2^{(\log_2 S_\ell + A)_+} \;\le\; 2^{A_+} C_0 \Big(\tfrac{L}{1-\beta} + L^\beta \Big) + L \;=\; O(L),$$

and $\sum_\ell c\, 2^{r_\ell^*/c} = O(L)$ for every kernel shape $c \ge 1$; with the r-independent address commit $\Theta(L)$, the Shout total is $\Theta(L)$;

(iii) the bound is sharp in β: at $\beta = 1$ the same sum is $O(L \log L)$ and at $\beta > 1$ only $O(L^\beta)$ — the $\beta < 1$ hypothesis is the one Corollary 1 already carries.

Proof. The envelope is the display of Corollary 1 at $k = L$: $S_\ell = \prod_{\ell<j\le L}(1 + c_j/\sigma_j) \le C_0 (L/\ell)^\beta$. (i) follows by taking $\log_2$ and adding A. For (ii), $2^{(x)_+} \le 2^x + 1$ gives $\sum_\ell 2^{(\log_2 S_\ell + A)_+} \le 2^{A_+} C_0\, L^\beta \sum_{\ell\le L} \ell^{-\beta} + L$, and $\sum_{\ell\le L} \ell^{-\beta} \le \frac{L^{1-\beta}}{1-\beta} + 1$ by integral comparison; for the kernel shape c the optimum $r_\ell^* = \frac{c}{c+1}(\log_2 S_\ell + A)_+$ gives $2^{r_\ell^*/c} \le (2^{A_+} C_0)^{1/(c+1)} (L/\ell)^{\beta/(c+1)} + 1$ with $\beta/(c+1) < 1$, so the same comparison applies. The address term is $\Theta(L)$ by r-independence and each per-proof factor is ≥ 1, so the total is $\Theta(L)$. For (iii), $\sum_{\ell\le L} (L/\ell)^\beta$ is $L\, H_L = \Theta(L \log L)$ at $\beta = 1$ and $\Theta(L^\beta)$ for $\beta > 1$. Numerically verified (explicit constants, β-sharpness, and the envelope on the ten measured EXP-A profiles): `verify_alpha_boundary.py`. □

Proof of Theorem 2. The first bullet is Lemmas 3–4 with the bit-as-cost node cost $\tilde{K}(r) = r$: the optimum is $r_\ell^* = (\log_2 S_\ell + A)_+$. Under the relative-budget convention ($E^* = \Theta(\sigma_L)$, §5) the unclipped level is $A = \log_2(CL/E^*) = (1-\alpha)\log_2 L + O(1)$: for $\alpha < 1$ it is eventually positive, so no layer clips and $\sum_\ell r_\ell^* = \sum_\ell \log_2 S_\ell + LA = \Theta(L^{2-\alpha}) + O(L \log L) = \Theta(L^{2-\alpha})$, the matching

lower order coming from the $c_k = \Theta(1)$ hypothesis via Lemma 4; at $\alpha = 1$, $A = O(1)$ and a constant fraction of layers retains $\log_2 S_\ell > |A|$, giving $\Theta(L)$; for $\alpha > 1$, $A < 0$ eventually and $\sum_\ell r_\ell^* \le \sum_\ell (\log_2 S_\ell)_+ = o(L)$. Altogether $O(L^{\max(1,2-\alpha)})$, and $\Theta(L^{2-\alpha})$ for $\alpha \le 1$, down from L^2 at $\alpha = 0$.

For Shout, the allocation-relevant per-proof kernel is $\tilde{K}(r) = \kappa_{\text{addr}}\, d\, m + \kappa\, c\, 2^{r/c}$ (extracted from the Jolt Atlas source, EXP-4d, §8.5). The address term $\kappa_{\text{addr}} dm$ is r-independent and proportional to the number of operations in layer ℓ, so summed over the L layers it is $\Theta(L)$ irrespective of the fidelities. The only r-dependent charge is the per-proof table-MLE evaluation $\kappa c\, 2^{r_\ell/c}$ (dense $c = 1$), one per layer. Two routes bound its depth total. (1) Allocation route: under the linear floor with $\beta = \bar{c}/a < 1$ (Corollary 1's hypotheses; $\alpha \ge 1$ is measured cross-model, EXP-A, §8.2), Lemma 5 gives $\sum_\ell c\, 2^{r_\ell^*/c} = O(L)$ even at the boundary $\alpha = 1$, where the early-layer r_ℓ^* is not bounded — no $\alpha > 1$ is invoked. (2) Deployed-fidelity route (Regime 3): r_ℓ is pinned to the deployed value (near the 16-bit operating point), so the table term is an $O(1)$ factor per layer by fiat; what requires proof is that the fixed fidelity stays sufficient, which is Corollary 1's depth-bounded normalized discrepancy under the same hypotheses, while the σ-floor (Proposition 1) rules out the σ-collapse that would force r_ℓ upward. Either route leaves the depth scaling $\Theta(L)$. Absent the floor, a σ-collapse (Regime 2) drives r_ℓ up without bound and detonates $2^{r_\ell/c}$ into the build wall (EXP-4a): the Shout-language form of the cost catastrophe, and the precise sense in which the floor is load-bearing. (Per-layer boundedness of $\log_2 S_\ell$ as depth grows would need activation growth $\alpha > 1$ — still a hypothesis: Theorem 1's full-alignment rate supplies only $\alpha \ge 1$, i.e. $\sigma_k = \Omega(k)$, and EXP-A's deep-half exponents (≥ 1.46 on 10/10 models) support but do not prove it, the fit windows being finite — via the convergent tail $\sum_{k>\ell} \sigma_k^{-1} = O(\ell^{1-\alpha})$ of Lemma 4. The headline no longer rests on it: at the measured $\alpha \ge 1$ boundary Lemma 5 closes the summed table term, and the bit-as-cost reading is $\sum_\ell r_\ell^* = \Theta(L)$ at $\alpha = 1$ by Lemma 4.) □

A.4 The attention degree gap (Theorem 3)

Lemma 6 (affordable degree)**.** *For $\mathcal{C}$ to contract each query in $o(N_{\text{seq}})$ work its shared moment features must number $m = \sum_{t \le p} \binom{d_{head}+t-1}{t} = \binom{d_{head}+p}{p} = o(N_{\text{seq}})$. This holds iff $p \le p_{\text{aff}}(d_{head}, N_{\text{seq}})$, and $p_{\text{aff}} = \Theta(\log_{d_{head}} N_{\text{seq}})$ throughout the regime $p_{\text{aff}} \le d_{head}^{1-\delta}$ (any fixed $\delta > 0$) — at $d_{head} = 64$ this covers every N_{seq} through $\sim 2^{48}$ and beyond.*

Proof. Per query, $\mathcal{C}$ contracts $q_i^{\otimes t}$ against the symmetric moment tensors $T_t = \sum_j k_j^{\otimes t}$, at cost $\sum_{t \le p} \dim T_t = \sum_{t \le p} \binom{d_{\text{head}}+t-1}{t} = \binom{d_{\text{head}}+p}{p}$ (hockey-stick identity; building the T_t costs $N_{\text{seq}} \cdot m$, the same order). Writing $d = d_{\text{head}}$, the standard bounds $\left(\frac{d+p}{p}\right)^p \le \binom{d+p}{p} \le \left(\frac{e(d+p)}{p}\right)^p$ give $\log \binom{d+p}{p} = \Theta(p \log(d/p)) = \Theta(p \log d)$ whenever $p \le d^{1-\delta}$, so the affordability threshold $\binom{d+p}{p} < N_{\text{seq}}$ solves to $p_{\text{aff}} = \Theta(\log_d N_{\text{seq}})$ in that regime ($d = 64$: cumulative counts 65, 2145, 47905, 814385 at $p = 1, \ldots, 4$, so $p_{\text{aff}} = 1$ at $N_{\text{seq}} = 1024$, 2 at 2^{12}, 3 at 2^{16}). Beyond the regime ($N_{\text{seq}} \gtrsim d^{\,d^{1-\delta}}$, astronomically past practice) the count grows only polynomially in p, so more degrees become affordable; there the finite-N_{seq} threshold N_{seq}^* of Corollary 2 carries the argument instead. This is exact combinatorics, with no measured input. □

Lemma 7 (required degree)**.** *Let $p_{\text{req}}(R)$ be the least degree of a polynomial matching e^s across a de-sinked score range of width R to the deployed relative accuracy ε while staying positive. Then $p_{\text{req}}(R) = \Theta(R)$, independent of N_{seq}: a positive polynomial of degree $R + O(\sqrt{R \log(1/\varepsilon)})$ suffices, and every relatively ε-accurate polynomial (any $\varepsilon \le \frac{1}{3}$; positivity not even needed) has degree at least $\frac{R/2 - \ln 2}{\ln(3+2\sqrt{2})} \ge 0.283\, R - 1$. The deployed slope between these constants is an empirical calibration.*

Proof. Sufficiency and positivity. Expand about the left endpoint s_0 of the range. For $u = s - s_0 \in [0, R]$ the truncation $q_p(s) = e^{s_0} \sum_{k \le p} u^k/k!$ has relative error $|e^u - \sum_{k \le p} u^k/k!|/e^u = \Pr[\mathrm{Pois}(u) > p]$, which is monotone in u and hence at most $\Pr[\mathrm{Pois}(R) > p]$; a Chernoff bound makes this $\le \varepsilon$ once $p \ge R + \sqrt{2R \ln(1/\varepsilon)} + \ln(1/\varepsilon)$. Being a partial sum of e^u with $u \ge 0$, q_p has nonnegative coefficients and is therefore positive on the range, so it also meets the softmax positivity constraint. Hence $p_{\mathrm{req}}(R) \le R + O(\sqrt{R \log(1/\varepsilon)}) = O(R)$. *Linear growth.* Writing the range as $[c - \frac{R}{2}, c + \frac{R}{2}]$ and $t = (s - c)/(R/2)$, the target is $e^c \psi(t)$ with $\psi(t) = e^{at}$, $a = R/2$, whose Chebyshev expansion $e^{at} = I_0(a) + 2\sum_{n \ge 1} I_n(a) T_n(t)$ has degree-n coefficient $2I_n(a)$ (modified Bessel), with $I_n(a) = \frac{(a/2)^n}{n!}(1 + O(a^2/n))$. These coefficients begin their rapid decay only at $n = \Theta(a)$ (the surrogate $(a/2)^n/n!$ peaks at $n \approx a/2 = R/4$), so the numerical degree of e^{at} on $[-1, 1]$, the least p at which the truncation error reaches a fixed tolerance, is $\Theta(a) = \Theta(R)$, and the truncation is near-optimal for this entire function [31]. *Necessity.* The linear lower bound needs no positivity. Suppose $\deg q = p$ and $\sup_{u \in [0,R]} |q(u)e^{-u} - 1| \le \varepsilon \le \frac{1}{3}$. On $[0, R/2]$ this forces $|q| \le (1 + \varepsilon)e^{R/2}$, while at the right endpoint $q(R) \ge (1 - \varepsilon)e^R$. A degree-$p$ polynomial bounded by M on $[0, R/2]$ is bounded at R by $M\, T_p(3)$ (Chebyshev growth: $R \mapsto 3$ under the affine map of $[0, R/2]$ onto $[-1, 1]$), and $T_p(3) \le (3 + 2\sqrt{2})^p$. Hence $(3 + 2\sqrt{2})^p \ge \frac{1-\varepsilon}{1+\varepsilon} e^{R/2} \ge \frac{1}{2} e^{R/2}$, i.e. $p \ge \frac{R/2 - \ln 2}{\ln(3+2\sqrt{2})} \ge 0.283\, R - 1$. Relative — not merely absolute-at-the-top — accuracy is the operative requirement here: the bulk holds $N'_{\mathrm{seq}} - n_{\mathrm{eff}} \gg n_{\mathrm{eff}}$ small-weight keys, and an absolute slack $\varepsilon e^{s_{\max}}$ per key admits spurious bulk mass up to $\varepsilon N'_{\mathrm{seq}} e^{s_{\max}}$ against $Z \le n_{\mathrm{eff}} e^{s_{\max}}$ ($w_{\max} \ge 1/n_{\mathrm{eff}}$), so ε would have to shrink like $n_{\mathrm{eff}}/N'_{\mathrm{seq}}$; uniform relative accuracy (with positivity, which EXP-6's `nan`/perplexity thresholds probe) is the N_{seq}-free formulation. The exact slope of the deployed relative-plus-positivity requirement is an empirical calibration, $p_{\mathrm{req}} \approx 1.21\, R + 2.5$ ($R^2 = 0.995$, EXP-6), a value between the entire-function rate 1 and the interval rate $e/2$; it is the one measured input. The positivity route gives the same order independently: below the accuracy threshold the fitted approximant dips negative on the small-score tail (measured: 41% of fitted exp-values negative at $p = 2$), so positivity alone already forbids $p = O(1)$. □

Proof of Theorem 3. The class $\mathcal{C}$'s only error is the polynomial approximation of exp: the moment identity $\sum_j \sum_{t \le p} c_t s_{ij}^t = \sum_{t \le p} c_t \langle q_i^{\otimes t}, T_t \rangle$ is exact, since $s_{ij}^t = \langle q_i^{\otimes t}, k_j^{\otimes t} \rangle$ and $T_t = \sum_j k_j^{\otimes t}$ (exact algebraic identities, term by term; the 10^{-16} residual is only a sanity check — the score rank $\le d_{\mathrm{head}}$ is what keeps the T_t small, not what makes them exact). Hence proving attention with $o(N_{\mathrm{seq}}^2)$ prover arithmetic within $\mathcal{C}$ requires $p_{\mathrm{aff}} \ge p_{\mathrm{req}}$, i.e. by Lemmas 6–7 $\Theta(\log_{d_{\mathrm{head}}} N_{\mathrm{seq}}) \ge \Theta(R)$, equivalently $R = O(\log_{d_{\mathrm{head}}} N_{\mathrm{seq}})$. This implication is measurement-free; the sole empirical input is that honest attention violates it. On GPT-2 the de-sinked range $R_{\mathrm{full}} \approx 9.83$ ($R_{\mathrm{eff}} \approx 5.6$) forces $p_{\mathrm{req}} \approx 10$–$14$, while $d_{\mathrm{head}} = 64$ gives $p_{\mathrm{aff}} \le 3$ even at $N_{\mathrm{seq}} = 2^{16}$: a 10–14× violation at $N_{\mathrm{seq}} = 1024$, still 3–5× at $N_{\mathrm{seq}} = 2^{16}$, so $\mathcal{C}$ cannot prove honest-input attention in $o(N_{\mathrm{seq}}^2)$; the quadratic is tight against $\mathcal{C}$. The conclusion is robust, as $p_{\mathrm{req}}/p_{\mathrm{aff}} = \Theta(R/\log_{d_{\mathrm{head}}} N_{\mathrm{seq}})$: the proven lower bound of Lemma 7 forces the gap at $N_{\mathrm{seq}} = 1024$ for every $R > 4.9$ (at $N_{\mathrm{seq}} = 2^{16}$ the proven threshold is $R > 12.0$; the 3–5× figure there rides the calibrated slope $1.21R + 2.5$), and $R \ge 8.6$ is measured across seven models (EXP-7, EXP-B17), so only a degenerate near-uniform attention escapes. □

A.5 The content-agnostic rank gap (Corollary 2)

Lemma 8 (second-moment reduction)**.** *Let the runtime value rows of V be zero-mean, of common expected energy, and uncorrelated across keys—that is, $\mathbb{E}VV^\top = cI_{N_{\mathrm{seq}}}$ for some $c > 0$, as holds*

for any i.i.d. zero-mean rows—and independent of the score geometry. Then for every fixed $\widehat{K}$*,*

$$\frac{\mathbb{E}_V\|\widehat{K}V - KV\|_F^2}{\mathbb{E}_V\|KV\|_F^2} = \frac{\|\widehat{K} - K\|_F^2}{\|K\|_F^2}.$$

Proof. For any fixed matrix A, $\mathbb{E}_V\|AV\|_F^2 = \mathbb{E}_V \mathrm{tr}(AVV^\top A^\top) = \mathrm{tr}\big(A(\mathbb{E}VV^\top)A^\top\big) = c\,\mathrm{tr}(AA^\top) = c\|A\|_F^2$. Applying this to $A = \widehat{K} - K$ and to $A = K$, the factor c cancels in the ratio. □

Remark (correlated values). For $\mathbb{E}VV^\top = \Sigma$ the same computation gives $\mathbb{E}_V\|AV\|_F^2 = \|A\Sigma^{1/2}\|_F^2$, so the identity persists in $\Sigma^{1/2}$-weighted norms and δ-accuracy forces $m \geq \mathrm{rank}_\delta(K\Sigma^{1/2})$: the rank bound on the principal value directions.

Proof of Corollary 2. A content-agnostic method fixes a query-independent per-key map β and forms the shared sketch $\Sigma = \sum_j \beta(k_j)v_j^\top$, returning $\widehat{\mathrm{num}}_i = \alpha(q_i)^\top\Sigma$; its effective kernel $\widehat{K}_{ij} = \alpha(q_i)^\top\beta(k_j)$ has rank $\leq m$ and cost $O(N_{\mathrm{seq}}\, m)$, sub-quadratic iff $m = o(N_{\mathrm{seq}})$. Because β sees neither the runtime values V nor the global K, Lemma 8 pins its error to the kernel rather than one instance: δ-accurate output needs $\|\widehat{K} - K\|_F \leq \delta\|K\|_F$. The best rank-$\leq m$ Frobenius approximation of K is its truncated SVD (Eckart–Young–Mirsky), whose error is $\big(\sum_{i>m}\sigma_i(K)^2\big)^{1/2}$; so a rank-$\leq m$ $\widehat{K}$ can be δ-accurate only if $m \geq \mathrm{rank}_\delta(K) := \min\{r : \sum_{i>r}\sigma_i(K)^2 \leq \delta^2\sum_i \sigma_i(K)^2\}$, and the optimal separable map is that truncated SVD. Now assume the spherical model: q_i, k_j on a common sphere, so that $K_{ij} = e^{\langle q_i, k_j\rangle}$ is a dot-product kernel there (real attention carries a norm spread; the model's reality checks are the numerical anchors below and the directly measured rank in the statement). By Funk–Hecke its Mercer eigenfunctions are the spherical harmonics: the degree-t eigenspace has dimension $\binom{d_{\mathrm{head}}+t-1}{t} - \binom{d_{\mathrm{head}}+t-3}{t-2}$ with a common eigenvalue, so the SVD is realized by polynomial features and no separable map improves on the harmonic expansion. Accumulating harmonics through degree p_{req} gives $\mathrm{rank}_\delta(K) = \Theta\big(\binom{d_{\mathrm{head}}+p_{\mathrm{req}}}{p_{\mathrm{req}}}\big)$ with the same $p_{\mathrm{req}} = \Theta(R)$ as Lemma 7 (the kernel inherits exp's approximation degree; the Frobenius-energy rank threshold here and the uniform-approximation degree of Lemma 7 are distinct quantities. We do not prove they coincide; we argue heuristically that both scale as $\Theta(R)$, since both are governed by the exponential kernel's coefficient decay across the range R—a matched-scaling heuristic, not a proved identity). Hence $m \geq F_\delta = \binom{d_{\mathrm{head}}+p_{\mathrm{req}}}{p_{\mathrm{req}}}$, which exceeds every practical N_{seq} (on GPT-2, 9.7×10^{10} at R_{eff} to 10^{15} at R_{full}); content-agnostic compression is thus impossible for all $N_{\mathrm{seq}} < N^*_{\mathrm{seq}} := F_\delta$, a finite-$N_{\mathrm{seq}}$ threshold rather than an asymptotic claim. Random-feature maps (Performer) realize the same harmonic spectrum in expectation, with added variance, and are therefore included. □

A.6 Cost equals bipartition rank (Propositions 3–4)

Lemma 9 (separation rank = bipartition rank)**.** *Fix a bit cut* $u \in \{0,1\}^a$*,* $v \in \{0,1\}^{r-a}$ *and let* $M_f[u,v] = T_f(u\,\|\,v)$ *be the rearranged value table. In the independent multilinear Lagrange bases* $\{\chi_u\}, \{\chi_v\}$*, the least number of terms* s *in a separated representation* $\tilde{f}(x_A, x_B) = \sum_{i\leq s} P_i(x_A)Q_i(x_B)$ *of the multilinear extension equals* $\mathrm{rank}_{\mathbb{F}}(M_f)$*; the ranks over the successive prefix cuts are the tensor-train ranks.*

Proof. The multilinear extension is $\tilde{f}(x_A, x_B) = \sum_{u,v} M_f[u,v]\,\chi_u(x_A)\chi_v(x_B)$, a bilinear form with matrix M_f in the bases $\{\chi_u\}, \{\chi_v\}$, which are linearly independent on the cube. A representation $\sum_{i\leq s} P_i(x_A)Q_i(x_B)$, restricted to the cube, expresses $M_f = \sum_{i\leq s}\hat{P}_i\hat{Q}_i^\top$ as a sum of s rank-one outer products, where $\hat{P}_i, \hat{Q}_i$ are the vectors of values of P_i, Q_i on the subcubes (restriction preserves the identity, so this holds whether or not the factors are multilinear); hence $s \geq \mathrm{rank}(M_f)$. Conversely,

any rank factorization of M_f, read in the bases $\{\chi_u\}, \{\chi_v\}$, assembles multilinear factors realizing $s = \operatorname{rank}(M_f)$. The least such s is therefore $\operatorname{rank}(M_f)$, and the ranks over the successive prefix cuts are the tensor-train ranks. □

Proof of Proposition 3. Cut the r address bits into $u \in \{0,1\}^a$ and $v \in \{0,1\}^{r-a}$ and form the $2^a \times 2^{r-a}$ value matrix $M_f[u,v] = T_f(u \,\|\, v)$. A separated representation $\tilde{f}(x_A, x_B) = \sum_{i \le s} P_i(x_A) Q_i(x_B)$ of the multilinear extension is, in the independent multilinear Lagrange bases $\{\chi_u\}, \{\chi_v\}$, a rank-s factorization of the bilinear form M_f; the least such s is $\operatorname{rank}_{\mathbb{F}}(M_f) =: \rho$ for that cut, and over the successive prefix cuts these ranks are the tensor-train ranks. A structured prover materializes exactly these ρ terms, so one node's per-proof table cost is $\tilde{\Theta}(\rho \cdot 2^{r/2})$ for a balanced two-way cut; at fidelity ε the operative quantity is the ε-approximate rank $\rho_\varepsilon = \#\{\sigma_i/\sigma_1 > \varepsilon\}$, and dense is the full-rank extreme $\rho = 2^{r/2}$. Structure sits low on this scale because a fixed-point value is additive across the cut, $\operatorname{val}(u \,\|\, v) = g(u) + h(v)$: an addition theorem gives an outer product ($\rho = 1$); a degree-δ polynomial factors through $\{g^0, \ldots, g^\delta\}$ ($\rho \le \delta + 1$); a holonomic function inherits $\rho_\varepsilon \le p(\varepsilon) + 1$ from its ODE-certified approximant. □

Proof of Proposition 4. Write $n = 2^{\lfloor r/2 \rfloor}$ and let N_f be the leading $n \times n$ submatrix of M_f. Since the rearrangement $M_f[u,v] = T_f(u \,\|\, v)$ is a bijection between matrix positions and table addresses, the n^2 entries of N_f are distinct table values ($n^2 \le 2^r$, so they exist). *Continuous entries.* $\det N_f$ is a polynomial in those entries that is not identically zero (it equals 1 on the identity table), so its zero set is a proper subvariety, of Lebesgue measure zero; hence N_f is invertible—and M_f has full rank $n = 2^{\lfloor r/2 \rfloor}$—with probability 1 for any continuous (e.g. Gaussian) entry distribution. *Uniform entries over a finite field $\mathbb{F}_q$.* Then N_f is a uniform random $n \times n$ matrix over $\mathbb{F}_q$. Constructing an invertible matrix column by column, the k-th column may be any vector outside the span of the previous $k-1$ independent columns, i.e. $q^n - q^{k-1}$ choices, so $|\mathrm{GL}_n(\mathbb{F}_q)| = \prod_{k=1}^n (q^n - q^{k-1})$ and

$$\Pr[\operatorname{rank} N_f = n] = \frac{\prod_{k=1}^n (q^n - q^{k-1})}{q^{n^2}} = \prod_{i=1}^n (1 - q^{-i}) \;\ge\; 1 - \sum_{i=1}^n q^{-i} \;>\; 1 - \frac{1}{q-1},$$

using $\prod_i (1 - a_i) \ge 1 - \sum_i a_i$ for $a_i \in [0,1]$. This bound is independent of n (hence of r), so a uniform table is full rank with probability $1 - O(1/|\mathbb{F}|)$; over a cryptographic field $|\mathbb{F}| \sim 2^{256}$ it is overwhelming. By Proposition 3 a generic table then needs $\rho = 2^{\lfloor r/2 \rfloor}$ committed polynomials — dense-level per-proof cost $\tilde{\Theta}(2^r)$, which no class of the model undercuts — so dense is forced: the generic table is its own witness. □

A.7 The differential-algebraic classification (Proposition 2)

Proof of Proposition 2. Fix the four-class model (dense; prefix–suffix; product/tensor; algebraic constraint) of §9. Each tier admits the claimed representation, hence the stated regime. An *algebraic* f (a low-degree relation $P(x, f) = 0$) is enforced on a witness with no value table, cost $\Theta(\deg \cdot m)$, wall-free and r-decoupled. An *addition-theorem* f ($e^{a+b} = e^a e^b$) factors as an outer product across any bit cut (Proposition 3, $\rho = 1$), giving the product path. A *holonomic* f (a linear ODE with polynomial coefficients: erf, hence $\mathrm{GELU}(x) = \frac{1}{2}x(1 + \operatorname{erf}(x/\sqrt{2}))$ by closure of D-finite functions under $+$ and polynomial $\times$), and more generally a *differentially-algebraic* f satisfying a polynomial ODE (tanh, sigmoid, SiLU, each obeying a first-order algebraic Riccati ODE, e.g. $\tanh' = 1 - \tanh^2$; not D-finite, as each has infinitely many complex poles, but ODE-certifiable all the same), admits an ODE-certified degree $p(\varepsilon)$ evaluation, table- and range-check-free, cost $\Theta(p \cdot m)$. *Periodicity/homogeneity* reduces the domain of whichever cell it feeds. The tiers are delimited by three classical facts: Weierstrass's classification [33] (a meromorphic function with

an algebraic addition theorem is rational in z, rational in $e^{\lambda z}$, or elliptic) bounds the addition-theorem tier; D-finiteness populates the holonomic tier; and Hölder's theorem [15] (Γ satisfies no algebraic differential equation) exhibits a function fitting none of the constructible classes. The forcing side is two-gated. Existence of dense-forced functions is unconditional (a generic table is full rank, Proposition 4). For a named anchor such as Γ, gate (I) certificate escape is provable (no product certificate by Weierstrass; not D-finite, as Γ has infinitely many poles $0, -1, -2, \ldots$; no algebraic differential equation by Hölder), whereas gate (II), an on-domain approximate-rank lower bound $\rho_\varepsilon = 2^{\Omega(r)}$, is an open frontier that abstract transcendence does not supply (Γ restricted to a pole-free interval is already compressible). The reduction that makes gate (II) concrete: M_f is a sampled additive kernel, $M_f[u, v] = K(g(u), h(v))$ with $K(x, y) = f(x+y)$, so its low-rank approximability is governed by f's analyticity on the evaluation range in the classical degenerate-kernel/hierarchical-matrix sense [14]: analytic on a Bernstein ellipse gives Chebyshev decay $C\rho_B^{-p}$ and $\rho_\varepsilon \sim \log(1/\varepsilon)$; merely C^k gives $\rho_\varepsilon \sim (1/\varepsilon)^{1/(k+1)}$ with a genuine dense crossover at tight fidelity; rough gives grid saturation [31]. The checked boundary cases confirm the ordering: Γ on the pole-free $[1, 5]$ has ε-rank 3 (compressible), a rough elementary function saturates at 123/128, and the SVD decay slopes reproduce the Bernstein ordering (erf -1.54/mode vs. C^1 -0.17). Every standard transformer activation carries a product, holonomic, or algebraic-differential certificate (each satisfies an algebraic differential equation, unlike Γ) and is on-domain analytic, so it lies in a constructible cell; the dense row is a boundary marker, not an obstruction transformers meet. □

A.8 The commitment floor (Corollary 3)

Proof of Corollary 3. By Proposition 5 the committed read-address object of m lookups has $m \cdot d$ one-hot nonzeros, fixed by the access multiset alone, so committing it is $\Omega(m)$ work irrespective of the table values. Function structure amortizes only the shared per-proof table $\tilde{f}$, never these m per-operation commitments; a sub-$\Omega(m)$ prover therefore requires either fewer operations (a model change) or a commitment scheme that exploits access-pattern compressibility (a different substrate). □

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